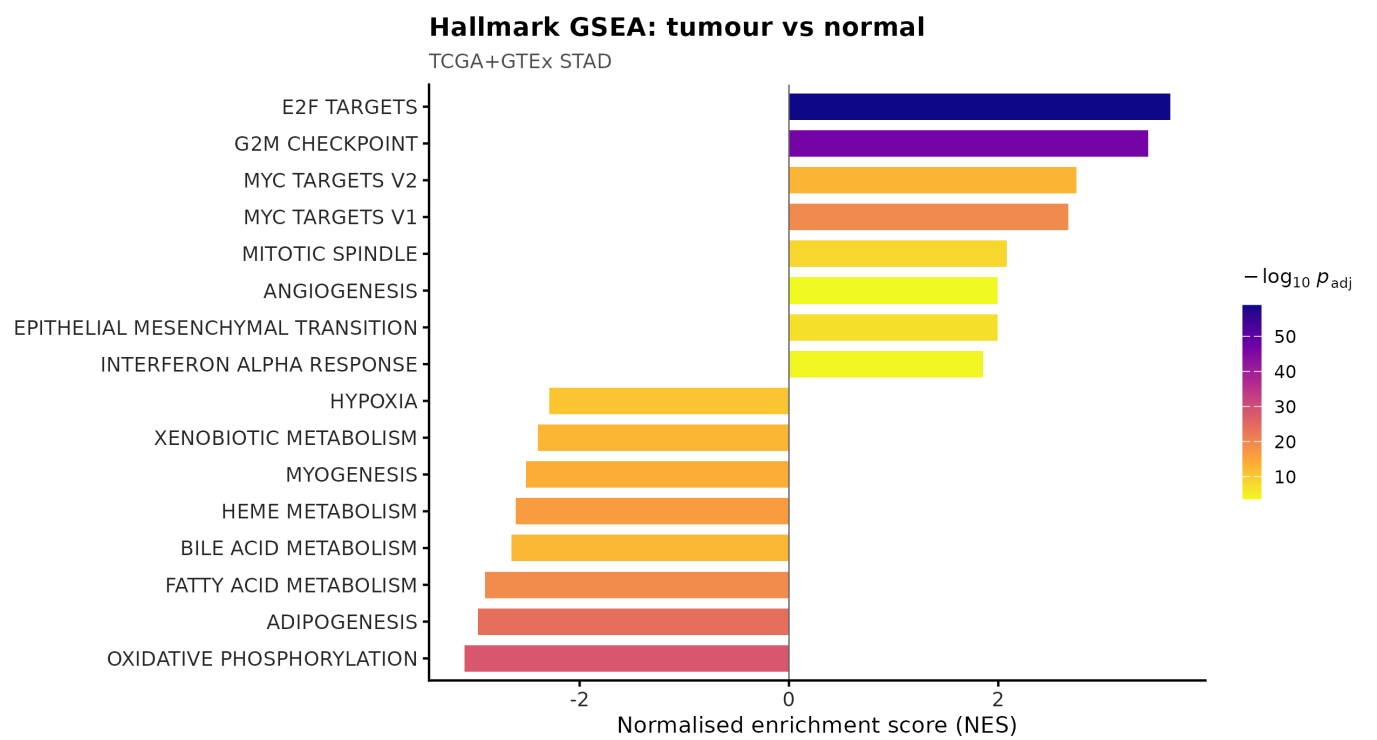


Gastric Cancer Multi-Omics

Complete figure set — every panel separated — one single-panel image per page

49 figures · generated 02 August 2026
each reproduced from its committed source file

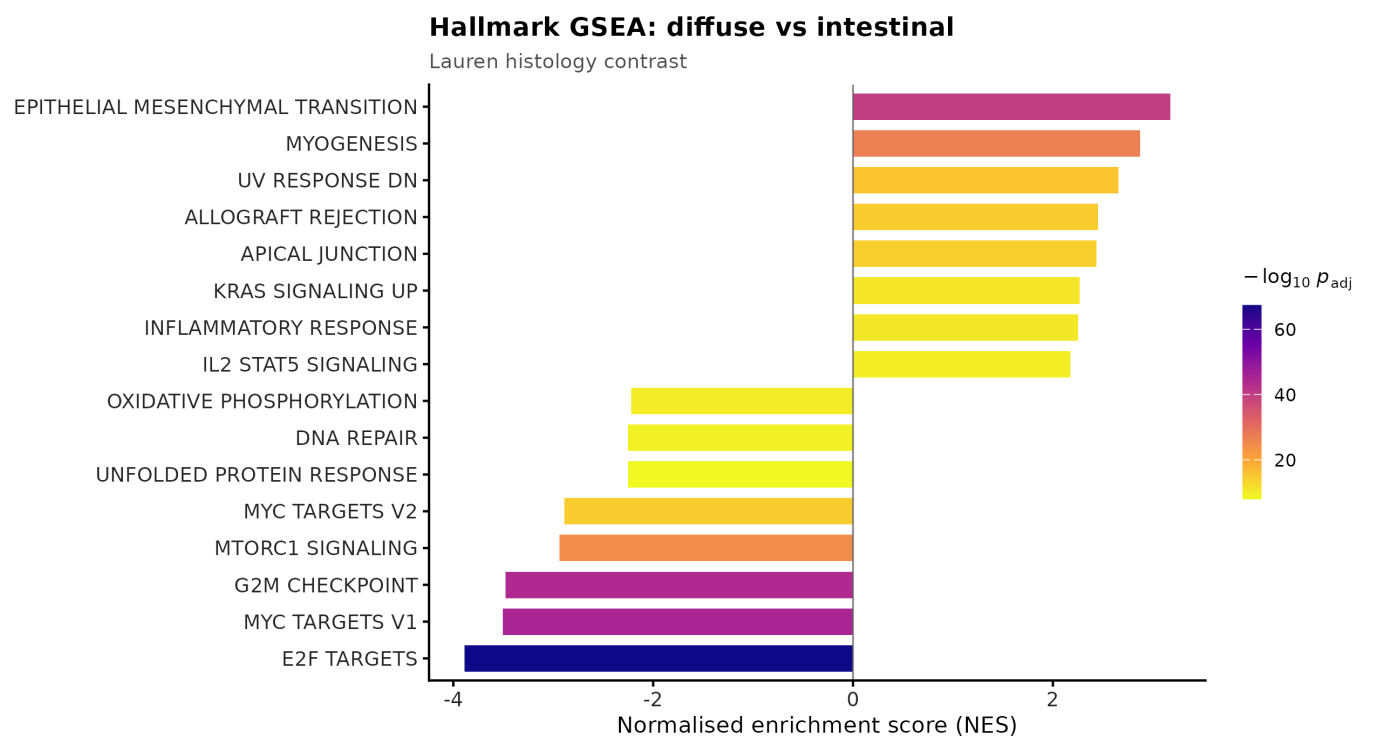
Figure 1



Hallmark gene-set enrichment, tumour versus normal (integrated TCGA+GTEx). Proliferation programmes are up (E2F targets NES +3.75; G2M checkpoint +3.64; MYC targets +2.67) and oxidative metabolism down (oxidative phosphorylation -2.54; fatty-acid metabolism -2.58).

[results/figures/panels/Fig1A.png](#)

Figure 2



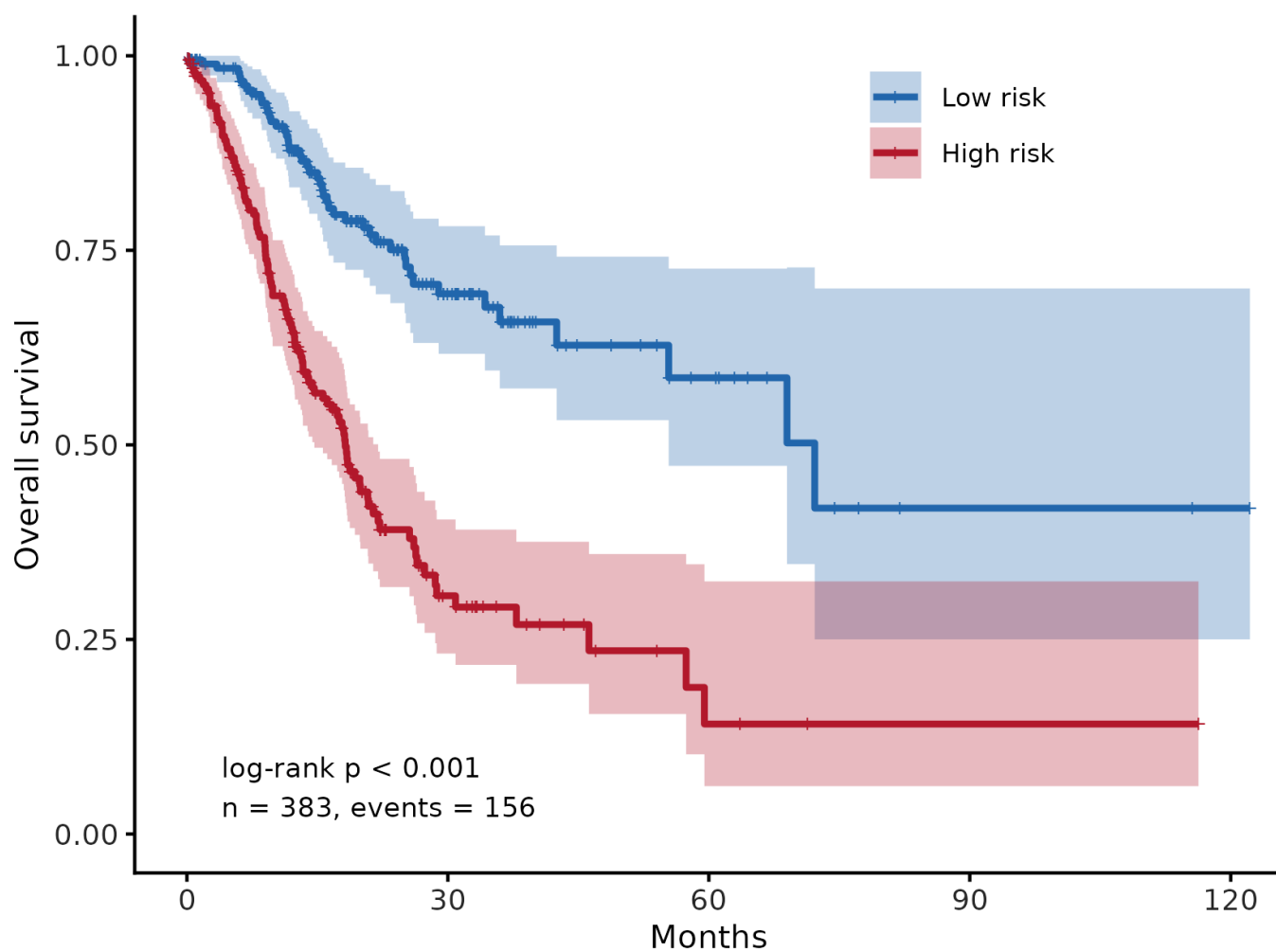
Hallmark gene-set enrichment, diffuse versus intestinal (Lauren contrast). Epithelial-mesenchymal transition is the top set in diffuse tumours (NES +3.17, adjusted $p = 1.6\text{e-}40$), recovering the classical histological distinction from expression alone.

[results/figures/panels/Fig1B.png](#)

Figure 3

25-gene risk score, TCGA-STAD

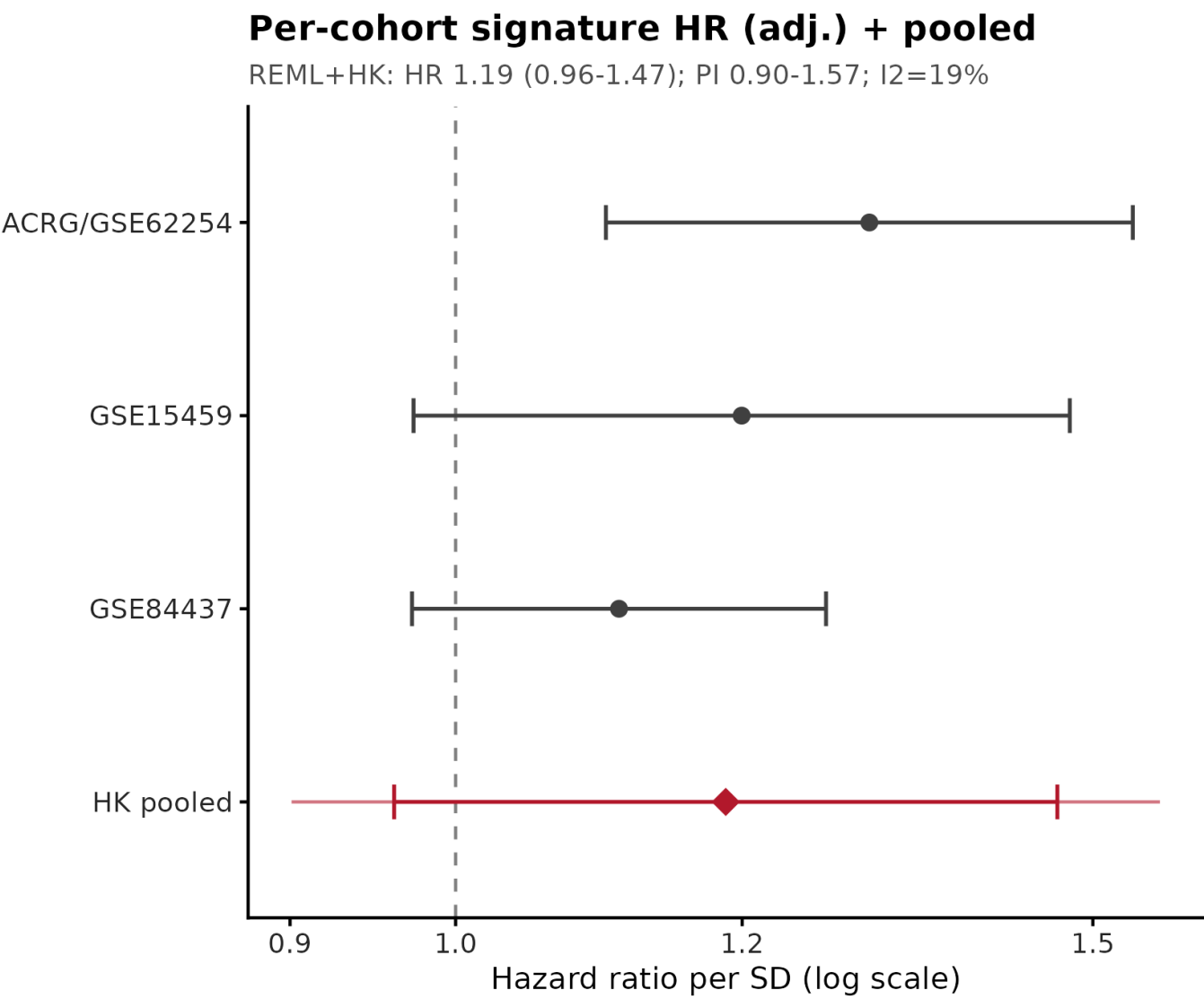
median split of z-scored signature



Kaplan-Meier stratification by the 25-gene signature in TCGA-STAD (training cohort, median split).

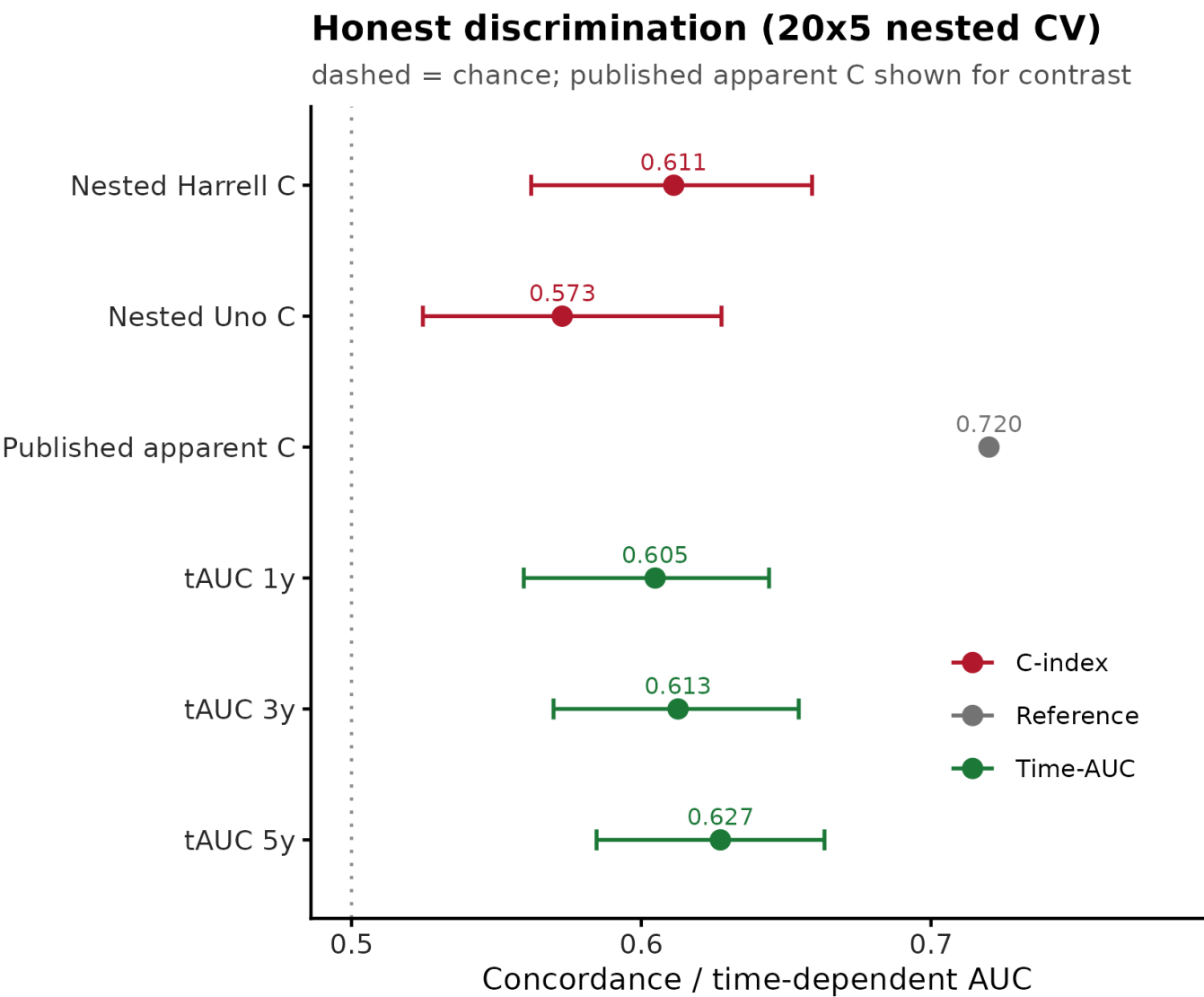
results/figures/panels/Fig2A.png

Figure 4



Per-cohort hazard ratios for the signature across training and external validation cohorts.
results/figures/panels/Fig2B.png

Figure 5



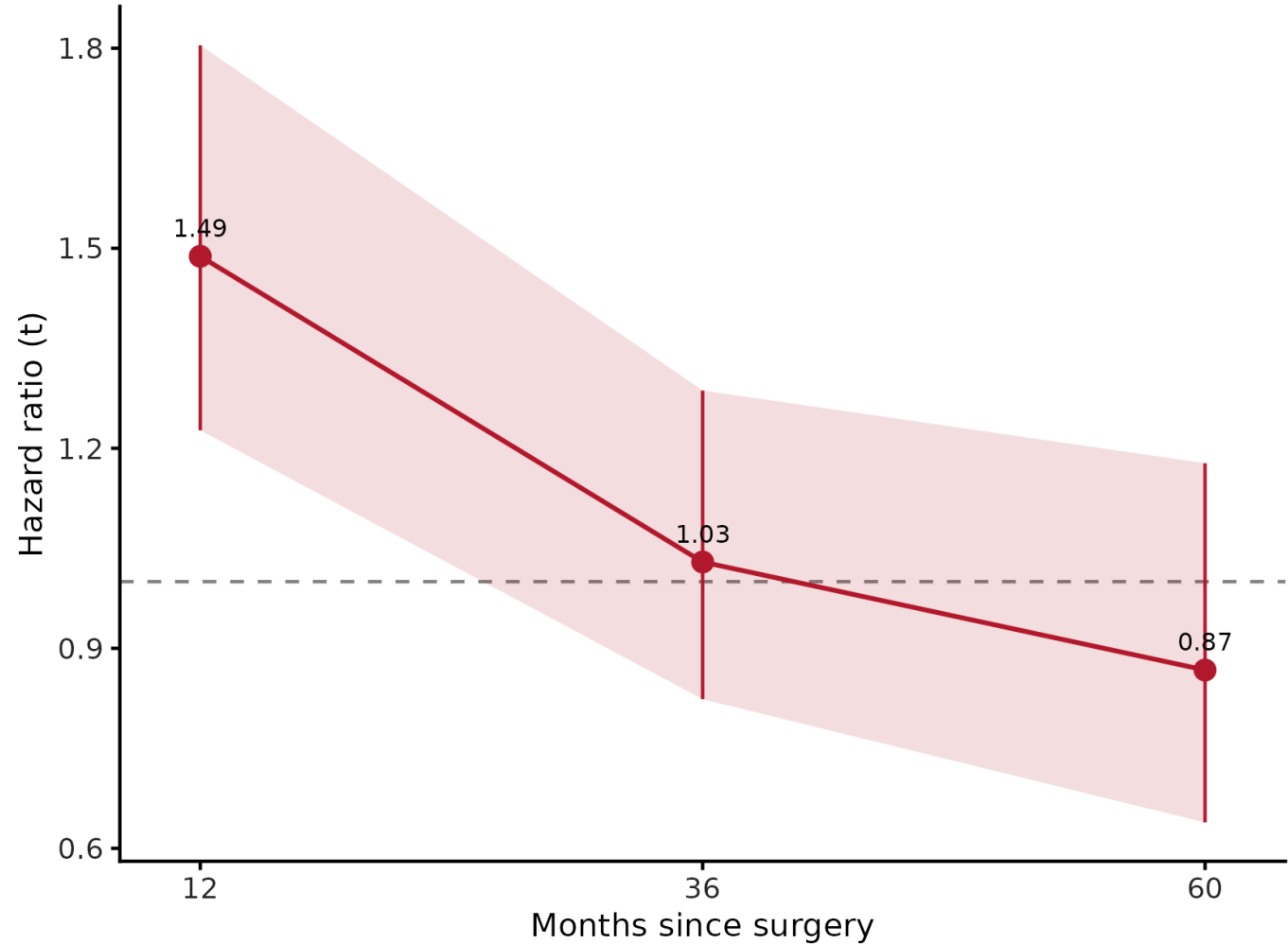
Internal performance estimated by nested cross-validation: Harrell C = 0.611 (95% CI 0.562-0.659), Uno C = 0.573, integrated Brier score 0.179. The apparent (leaky) C was 0.72.

results/figures/panels/Fig2C.png

Figure 6

Time-varying signature HR, ACRG

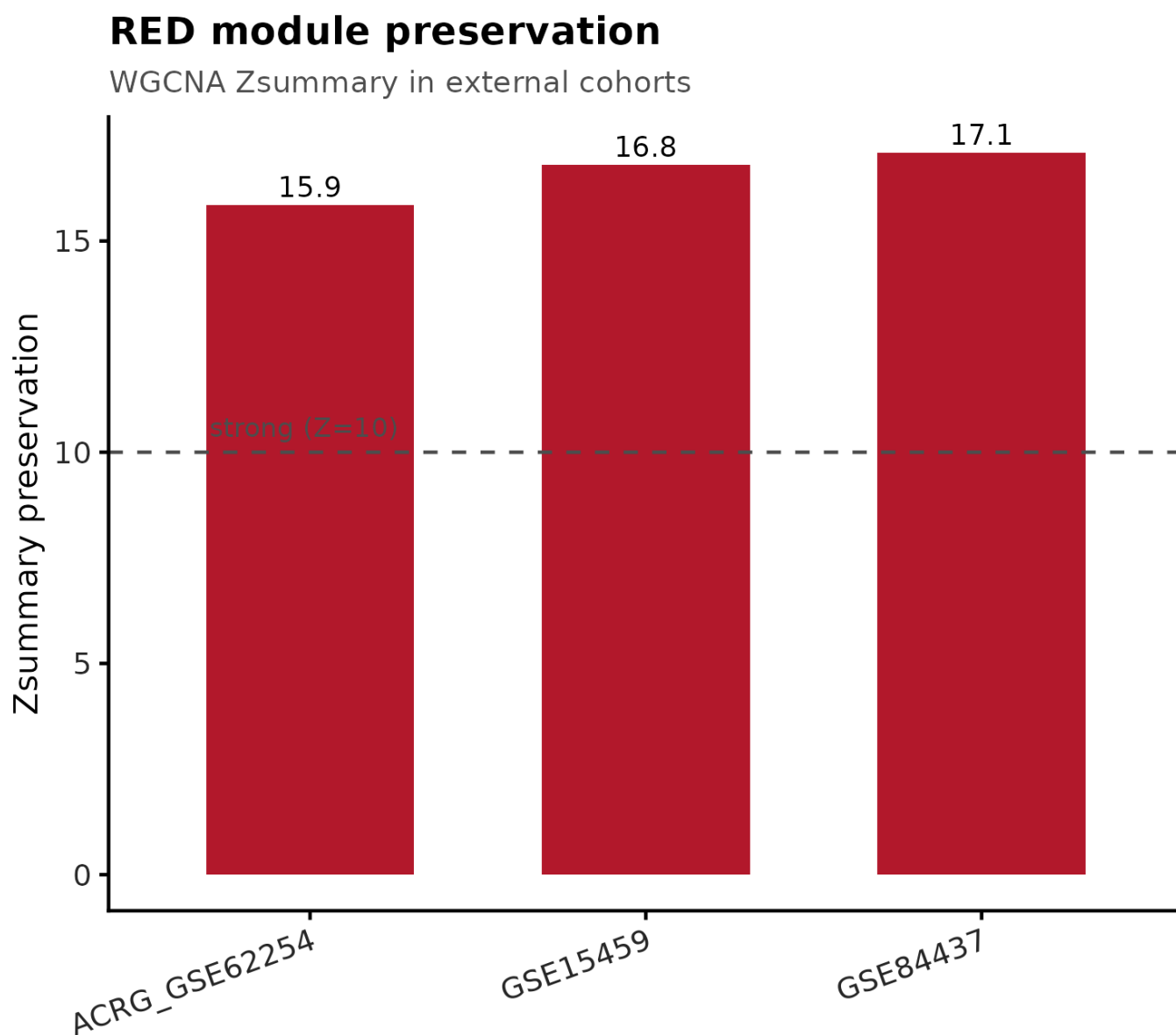
tt log-time Cox (age-adj, stage-stratified); early effect attenuates



Time-varying hazard ratio in ACRG. The signature is prognostic early (HR 1.49 at 12 months) and attenuates to null thereafter (1.03 at 36 months; 0.87 at 60 months); proportional hazards is violated (cox.zph p = 0.0018).

results/figures/panels/Fig2D.png

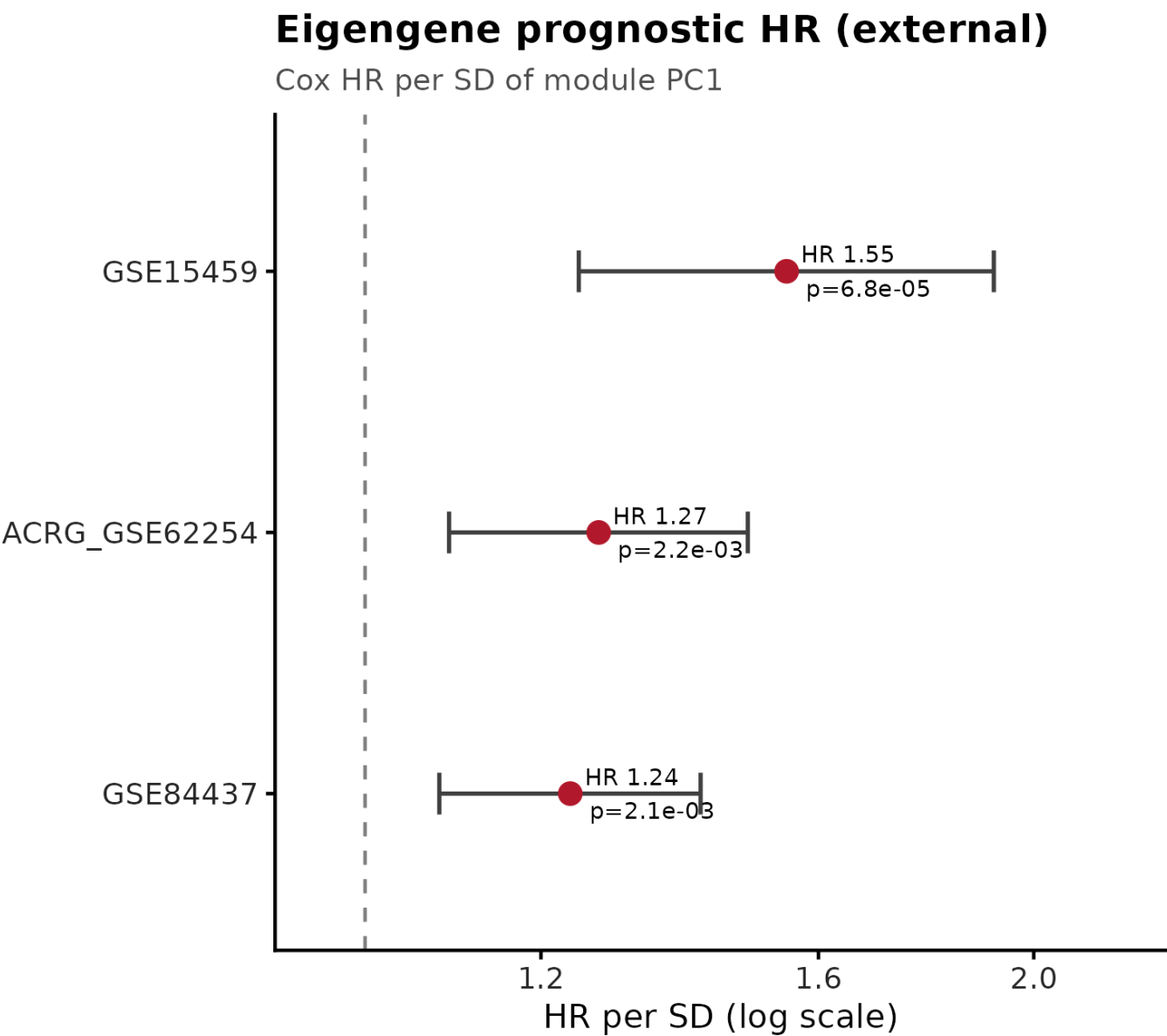
Figure 7



WGCNA module preservation of the stromal/fibroblast (red) module in three independent cohorts. Zsummary 15.9, 16.8 and 17.1 - all far above the $Z > 10$ strong-preservation threshold.

[results/figures/panels/Fig3A.png](#)

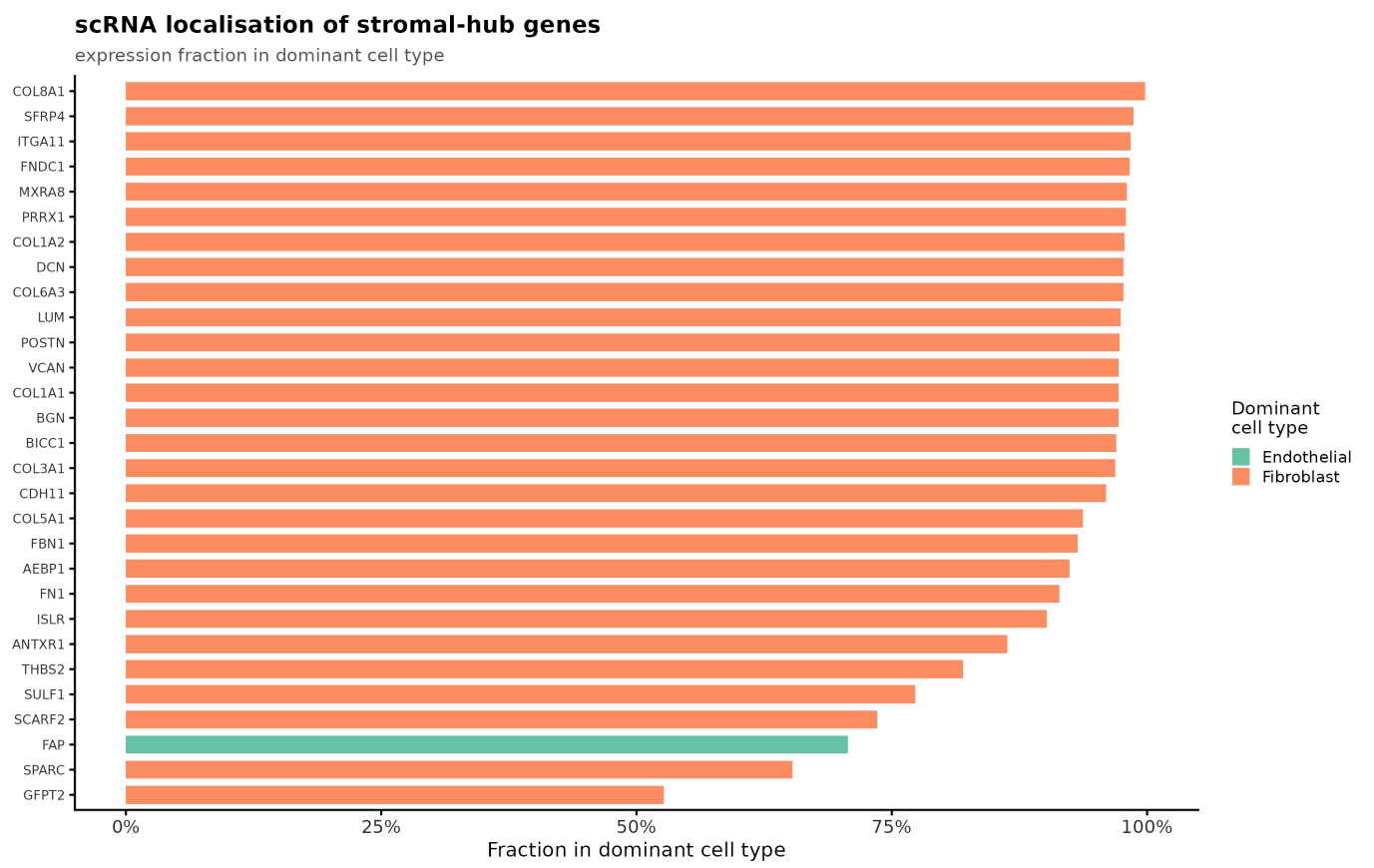
Figure 8



Module eigengene survival association in the same external cohorts (hazard ratio per standard deviation 1.24-1.55, all $p < 0.005$). The module replicates in GSE84437, where the 25-gene signature failed.

[results/figures/panels/Fig3B.png](#)

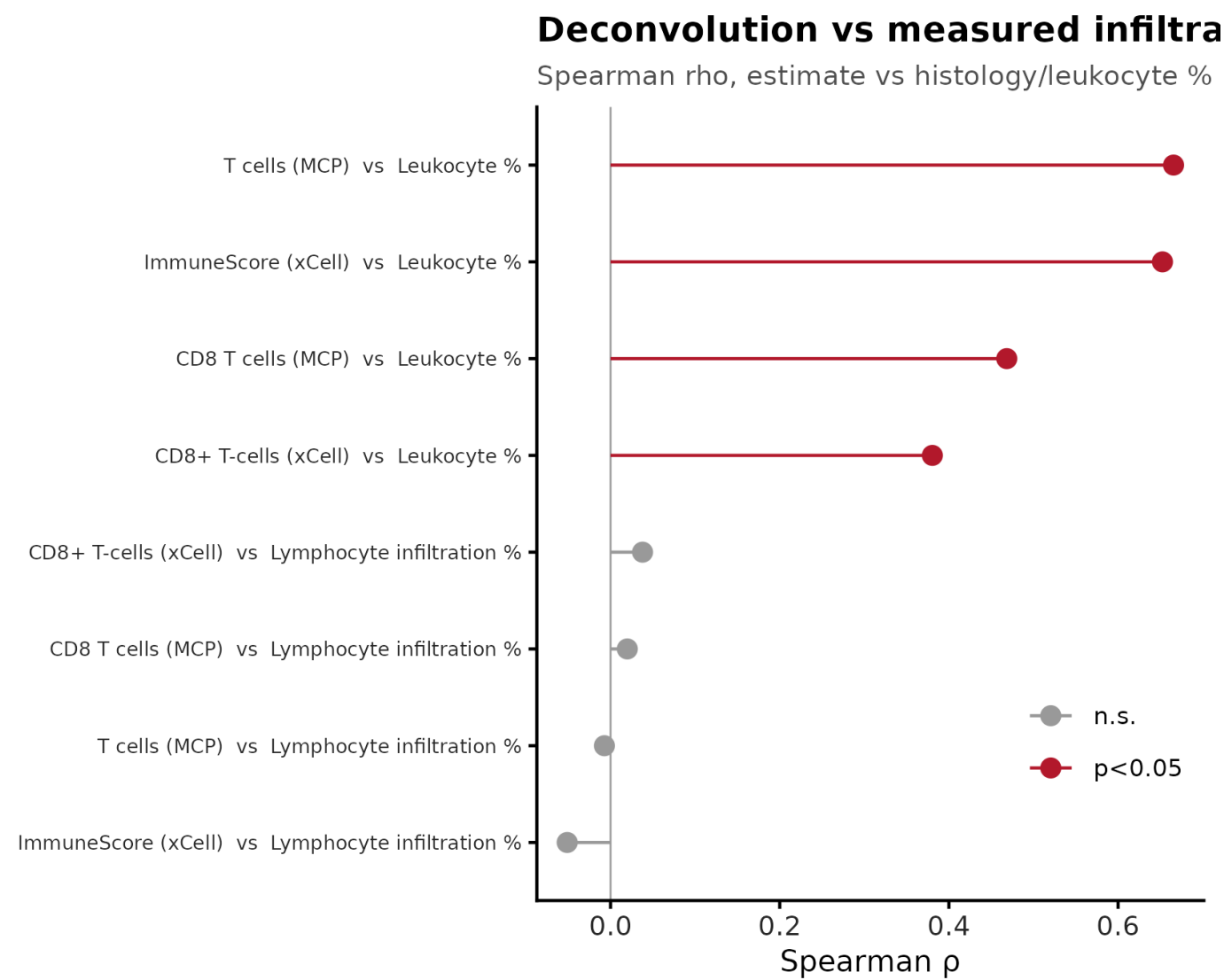
Figure 9



Single-cell localisation of the 29 stromal module hub genes: 28 of 29 are fibroblast-dominant (median fraction 0.97), the exception being FAP, which is endothelial-dominant here (0.71). Because six of these genes (DCN, LUM, COL1A1, COL1A2, FAP, COL3A1) were themselves used to annotate the fibroblast cluster, a non-circular re-test excluding them was also run: all 23 remaining hub genes stay fibroblast-dominant (median fraction 0.96).

results/figures/panels/Fig3C.png

Figure 10



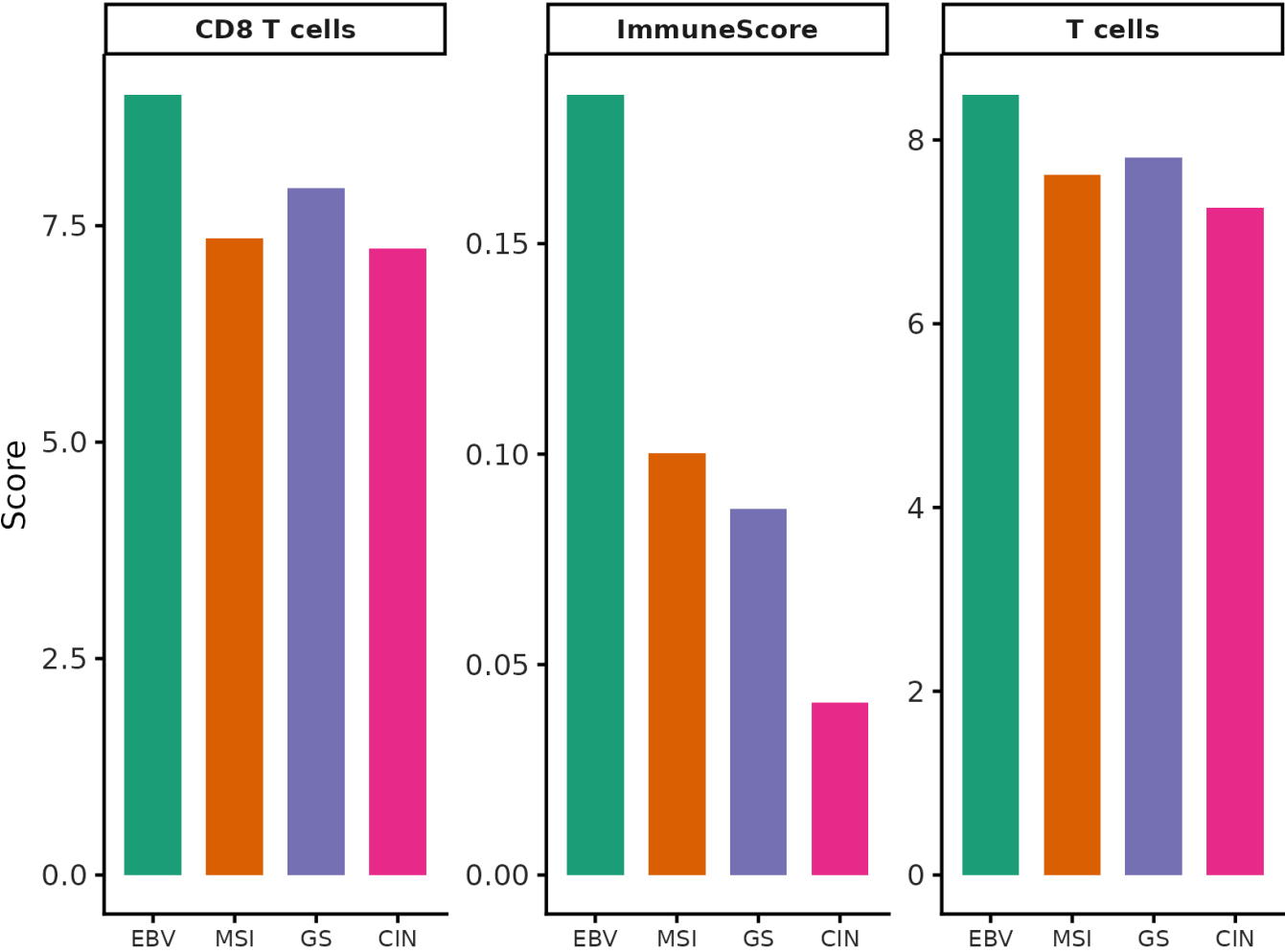
Immune deconvolution validated against pathologist-measured values. T-cell score versus measured leukocyte percentage: Spearman rho = 0.666, p = 3.6e-36, n = 272.

results/figures/panels/Fig4A.png

Figure 11

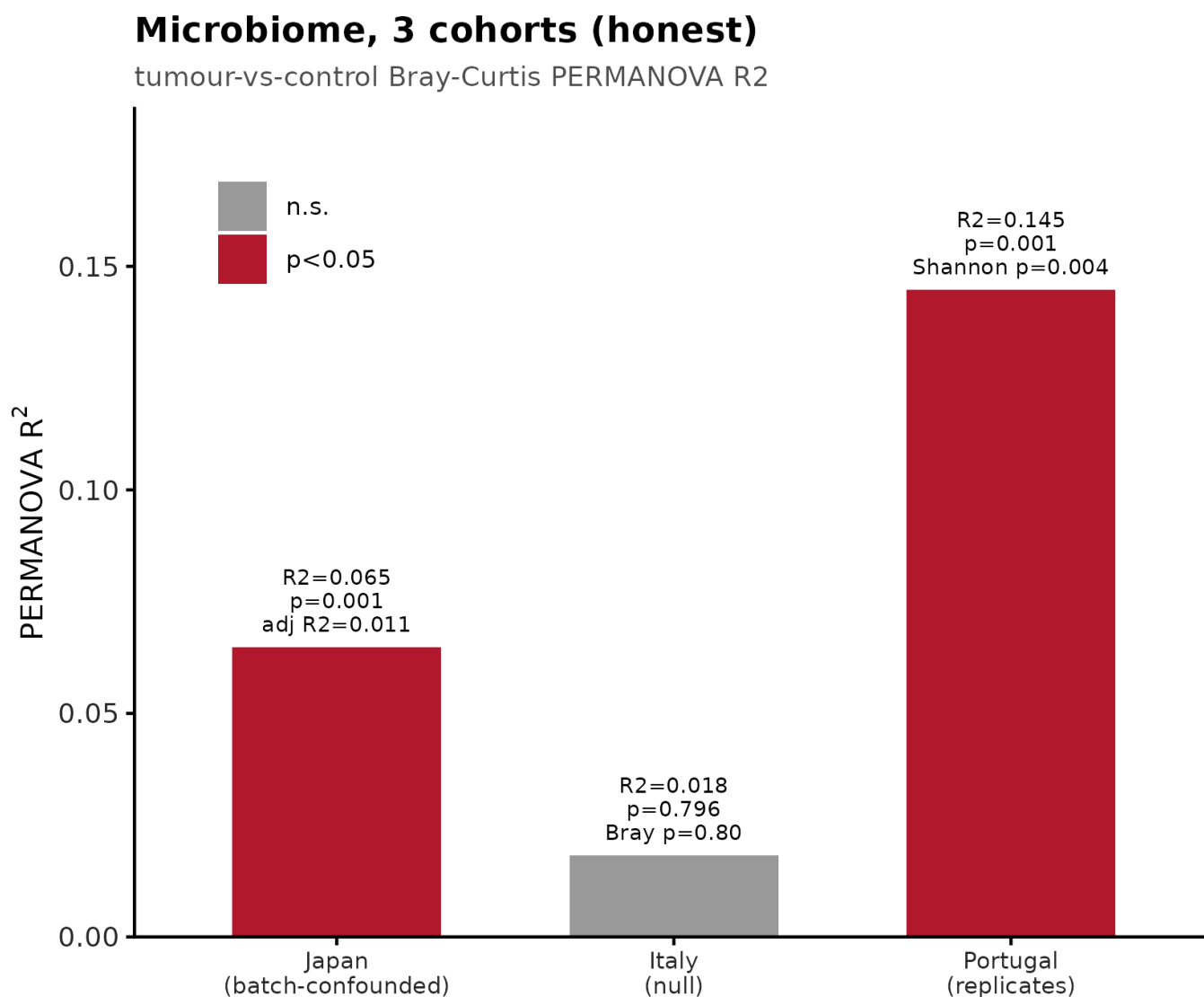
Immune scores by molecular subtype

all Kruskal-Wallis p_adj < 1e-6 (EBV/MSI immune-hot)



Immune infiltration by Lauren subtype.
results/figures/panels/Fig4B.png

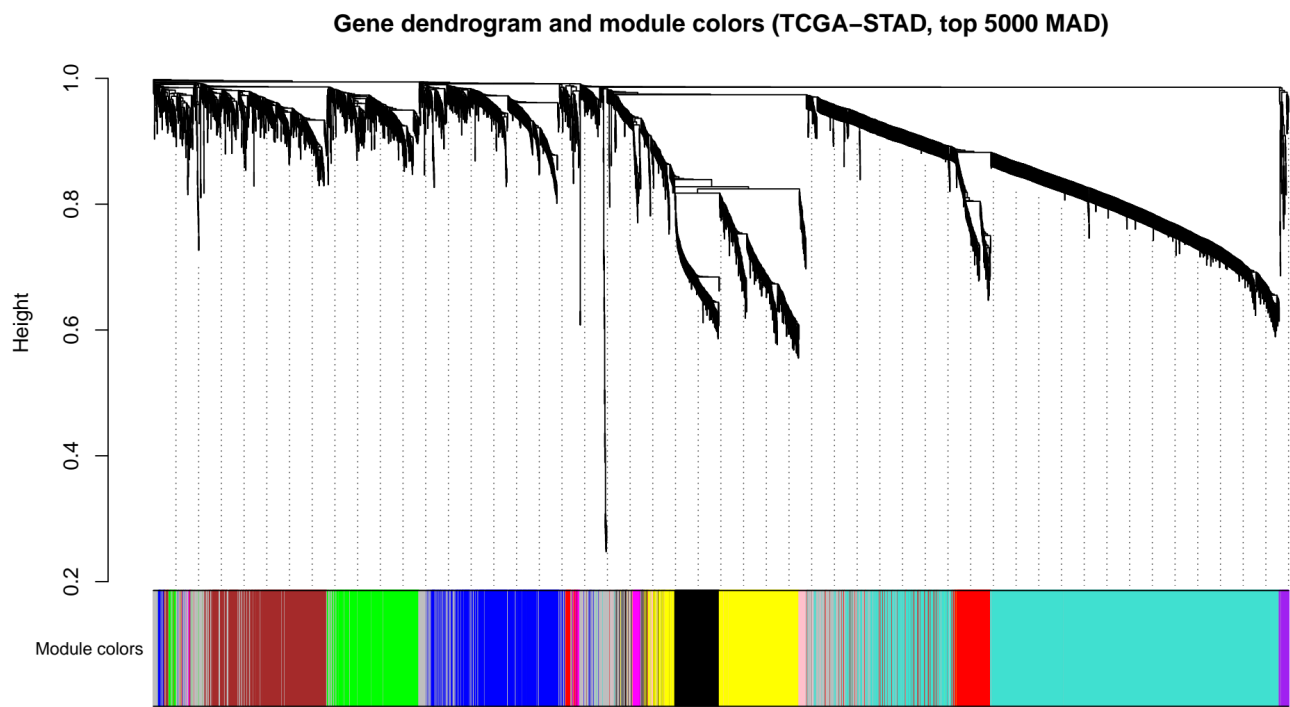
Figure 12



Three-cohort tissue-microbiome comparison. Japan: tumour signal batch-confounded (Bray-Curtis R² 0.065 to 0.011 after flowcell adjustment). Italy: null (R² = 0.018, p = 0.80). Portugal: reduced diversity replicates (Shannon p = 0.004; Bray R² = 0.145, p = 0.001).

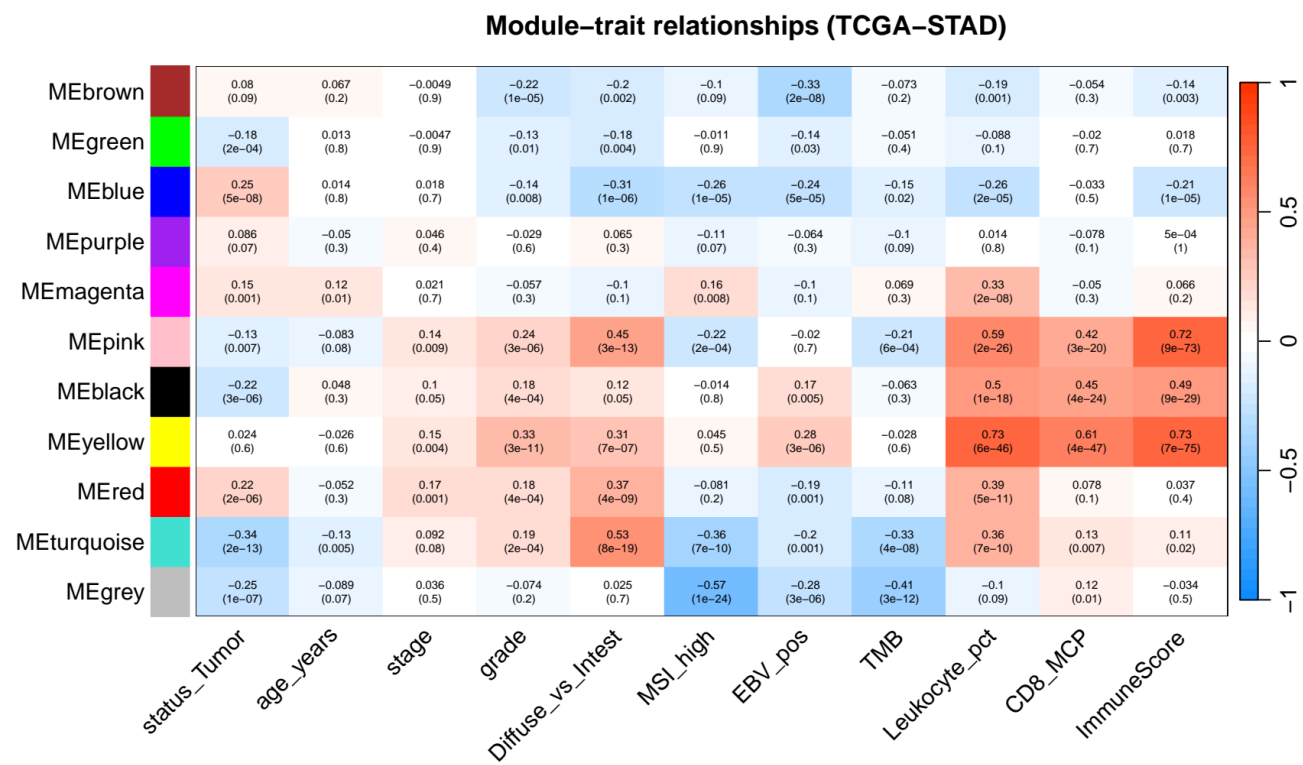
results/figures/panels/Fig4C.png

Figure 13



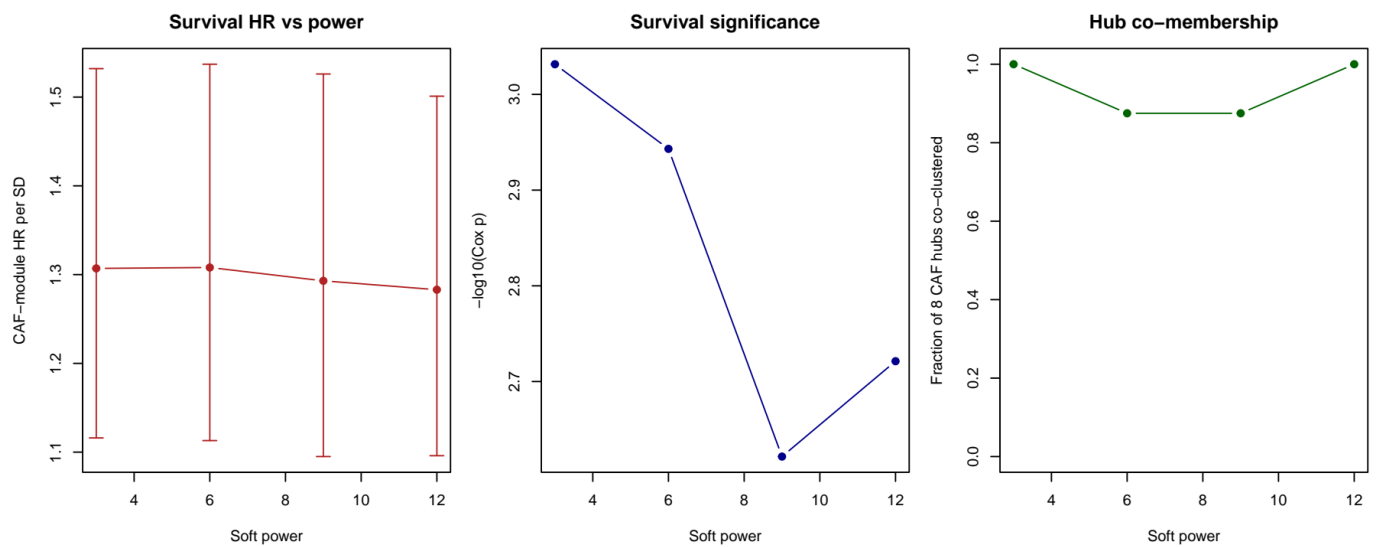
WGCNA gene dendrogram and module assignment (TCGA-STAD, top 5000 genes by median absolute deviation).
results/wgcna_real/wgcna_dendrogram.png

Figure 14



WGCNA module-trait relationship heatmap across clinical and molecular variables.
results/wgcna_real/wgcna_module_trait.png

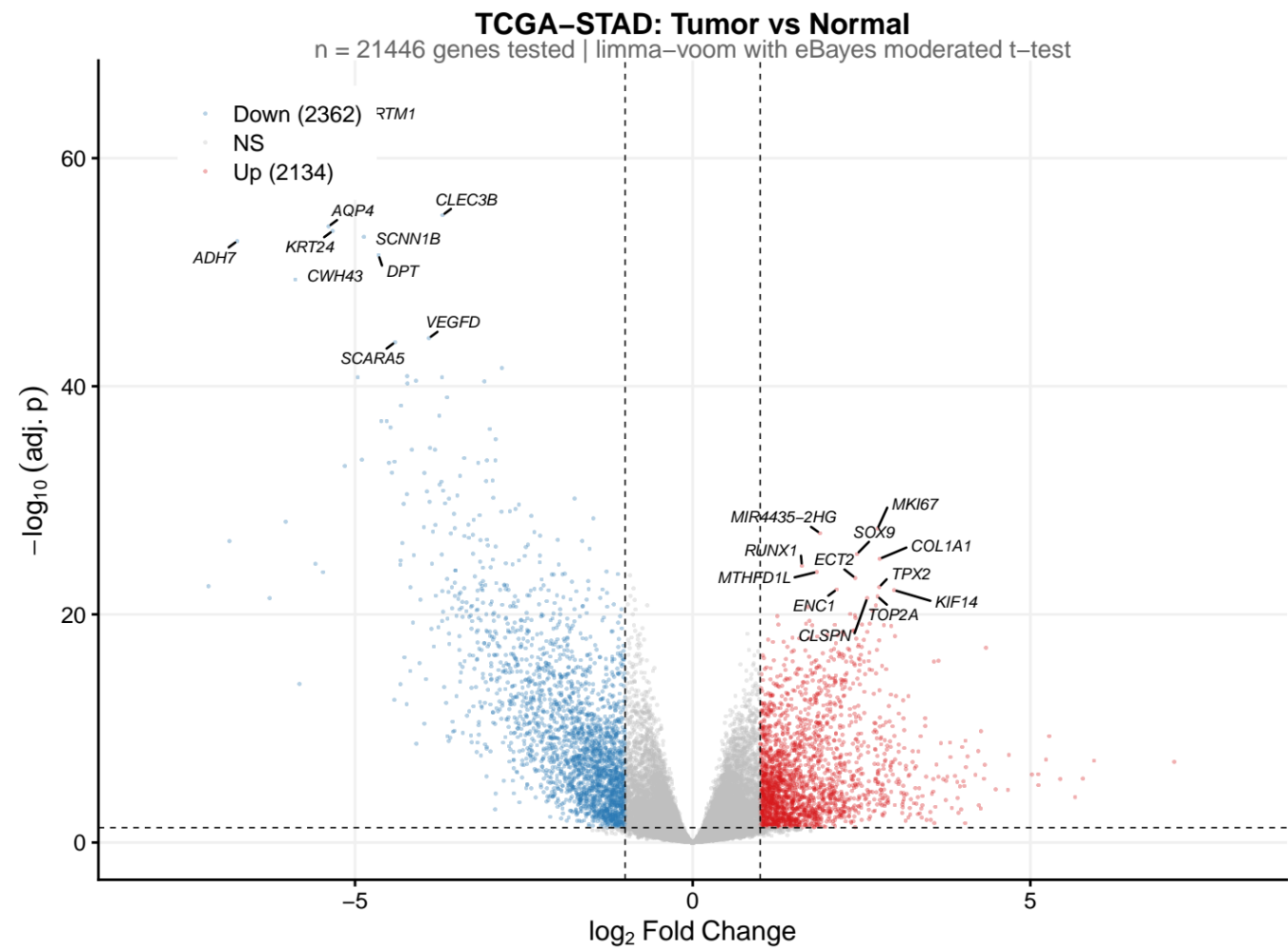
Figure 15



Soft-threshold power selection. Scale-free fit is near-flat across powers 3-12 (R^2 0.877-0.886); the prognostic module is invariant to the choice.

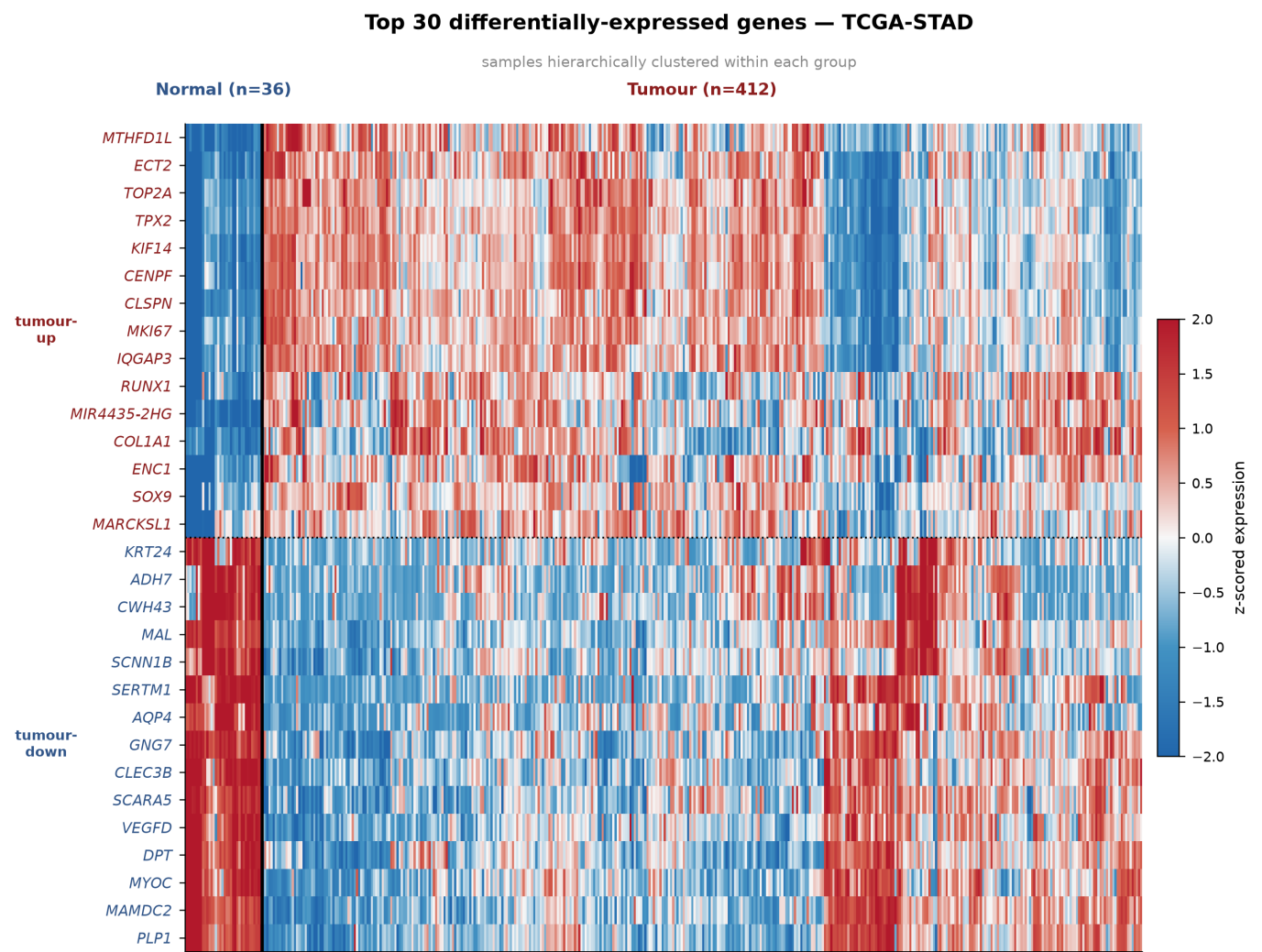
results/wgcna_real/wgcna_power.png

Figure 16



TCGA-only differential expression (limma-voom): 2,134 up and 2,362 down of 21,446 genes tested.
results/plots/transcriptome/deg_volcano.png

Figure 17

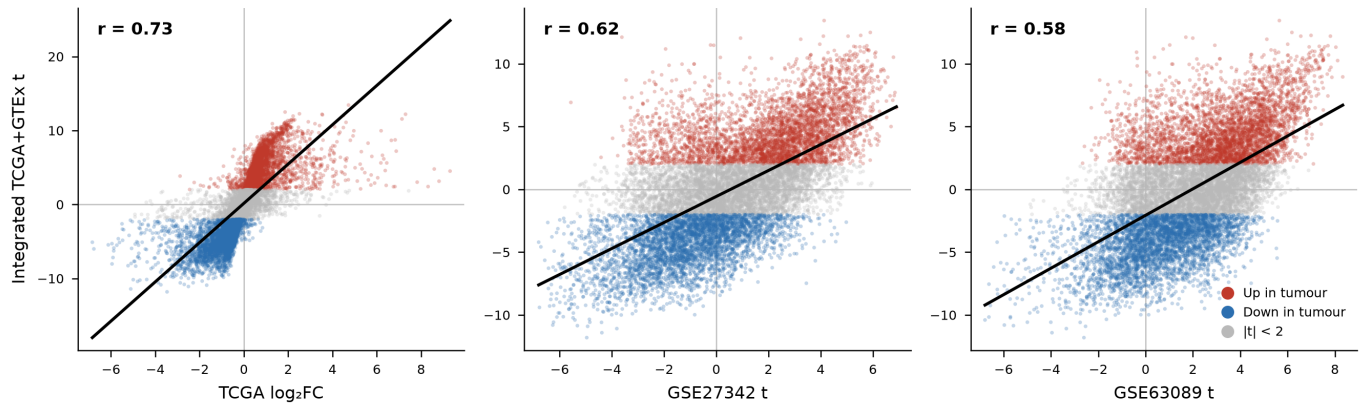


Top 30 differentially-expressed genes (15 up, 15 down) in 36 normal versus 412 tumour samples; colours are z-scored expression.

results/plots/transcriptome/deg_heatmap_top30_clean.png

Figure 18

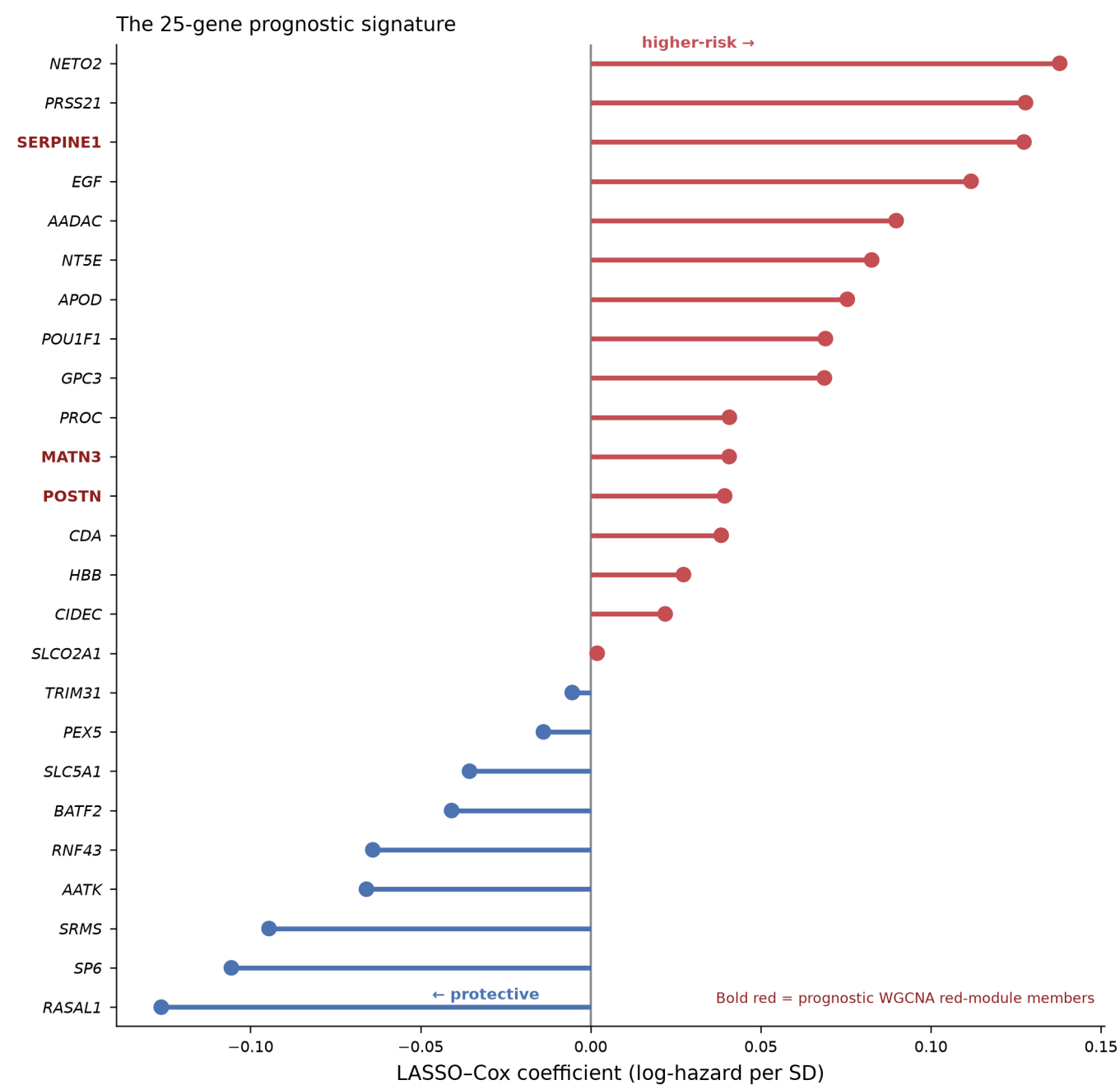
Integrated tumour-vs-normal ranking vs three independent cohorts (n=12,905 shared genes)



Integrated tumour-versus-normal ranking against three independent cohorts (n = 12,905 shared genes): TCGA $r = 0.73$, GSE27342 $r = 0.62$, GSE63089 $r = 0.58$.

[results/figures/deg_concordance_panel.png](#)

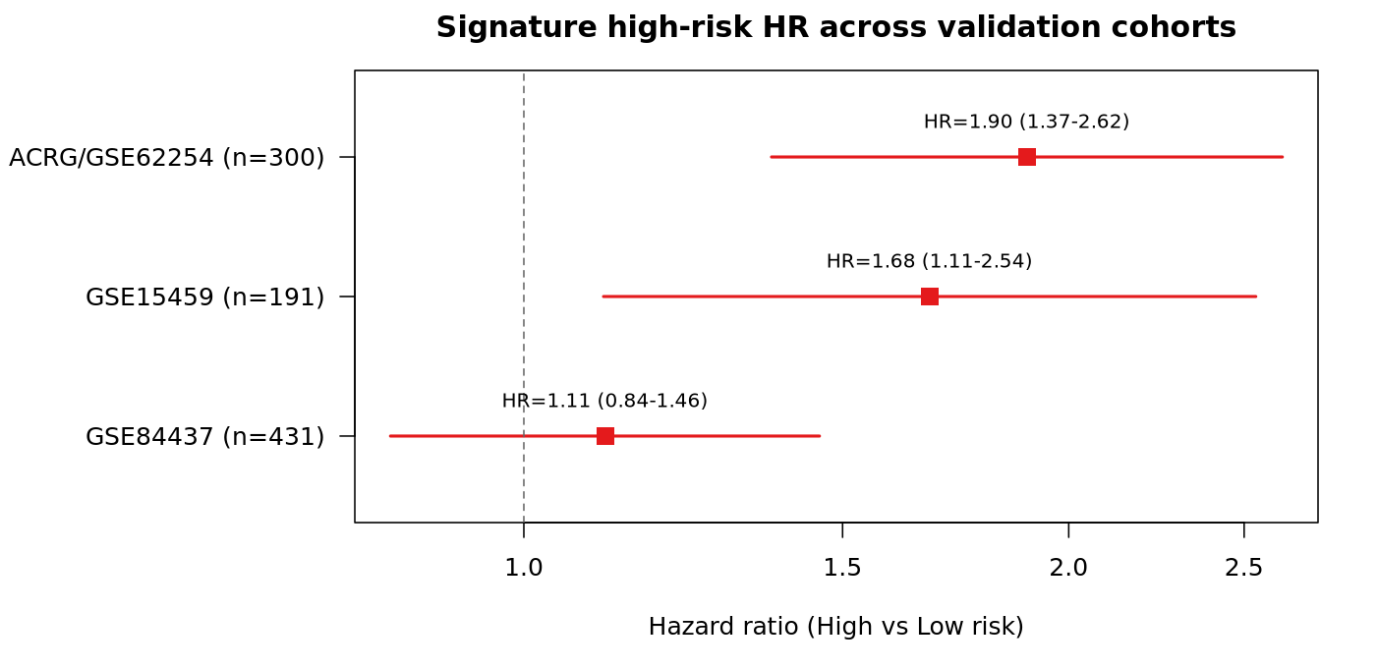
Figure 19



Standardised LASSO-Cox coefficients (log-hazard per standard deviation) for the 25 signature genes. Genes also in the prognostic WGCNA red module (SERPINE1, POSTN, MATN3) are highlighted.

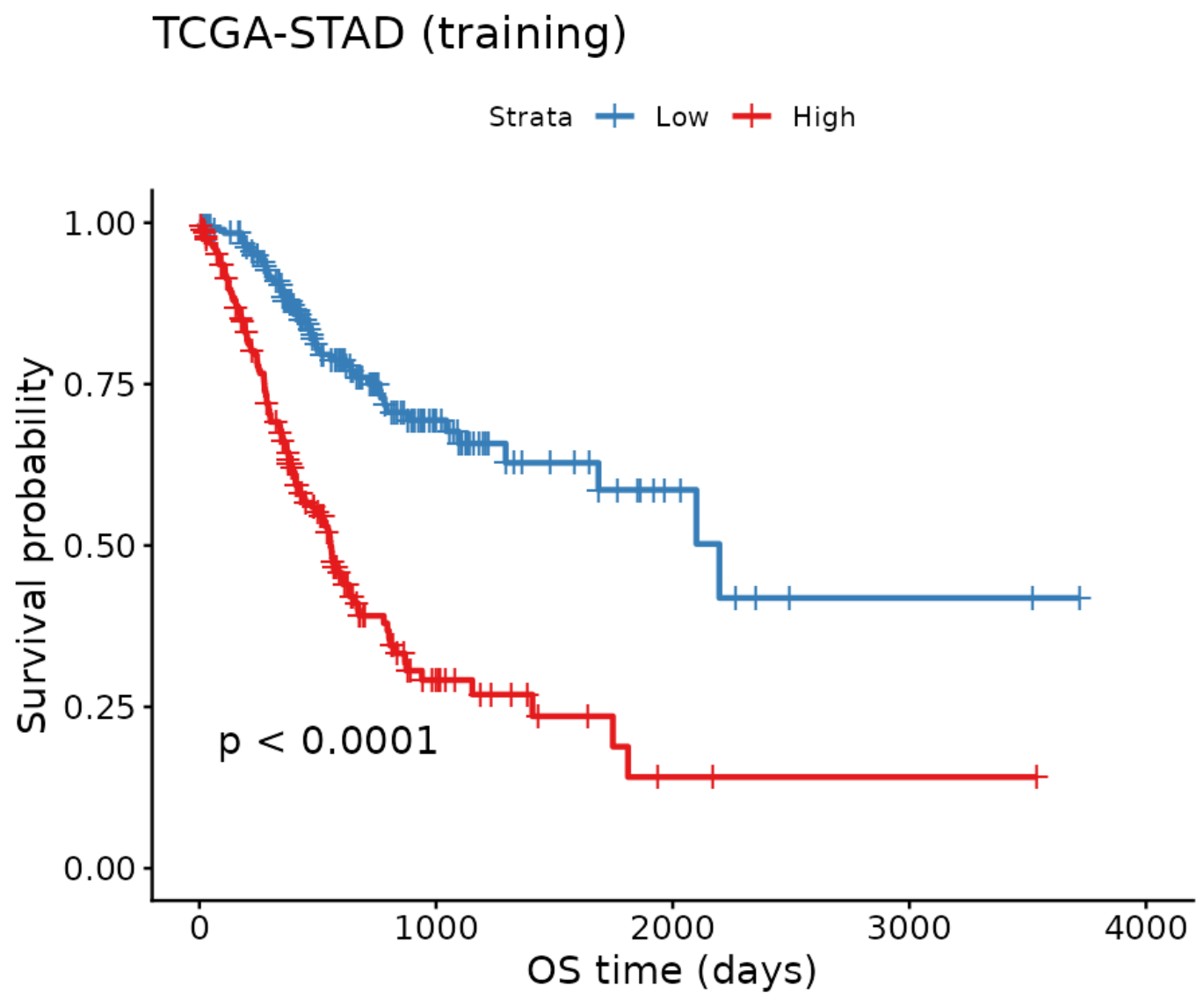
results/validation/signature_coefficients_clean.png

Figure 20



Per-cohort external-validation forest plot (high-versus-low hazard ratios, median split, unadjusted): ACRG 1.90 (1.37-2.62), GSE15459 1.68 (1.11-2.54), GSE84437 1.11 (0.84-1.46, null).
results/validation_multi/forest_HR.png

Figure 21

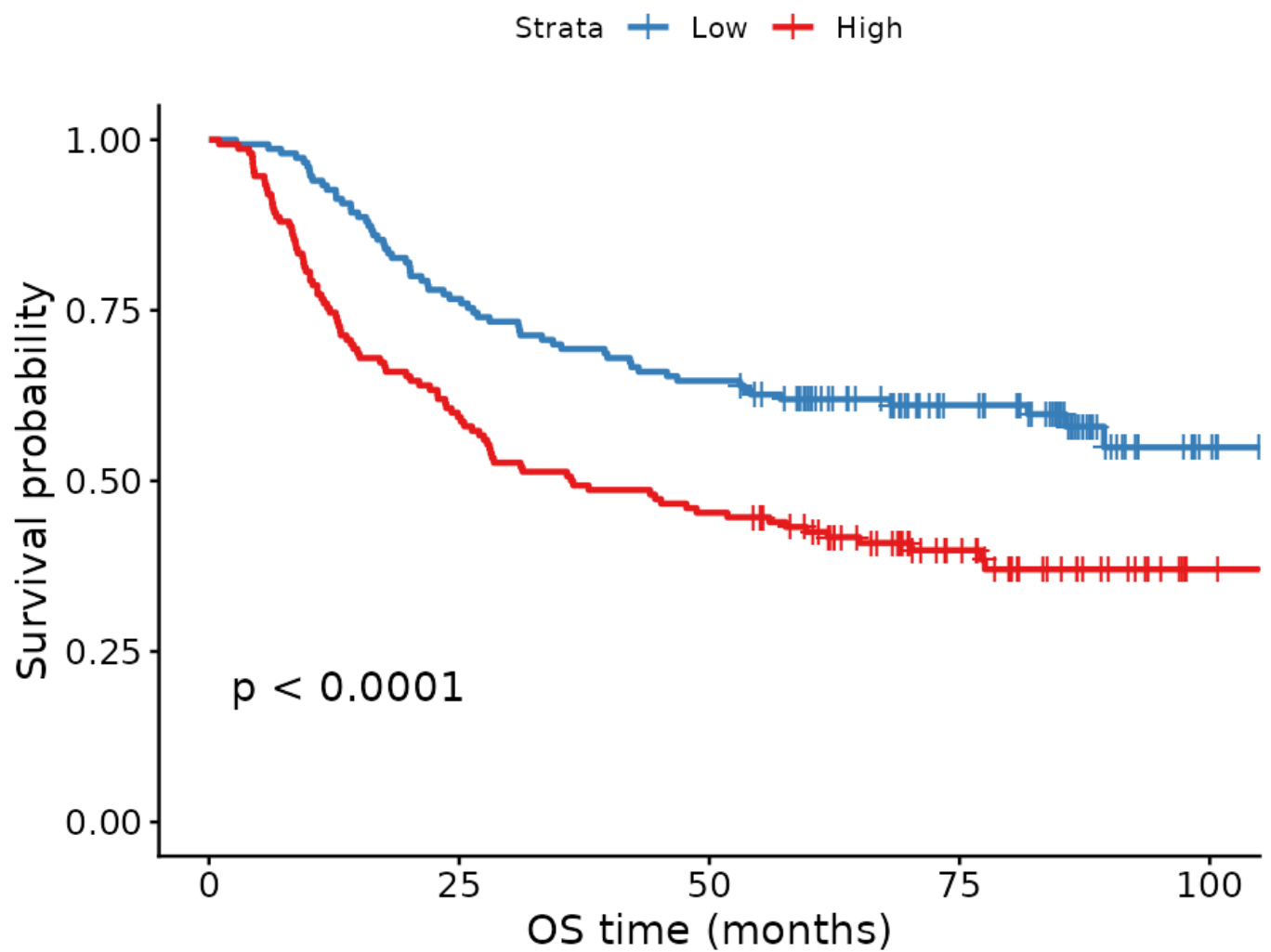


Kaplan-Meier stratification, TCGA-STAD (training).

[results/validation/KM_TCGA.png](#)

Figure 22

ACRG / GSE62254 (validation)

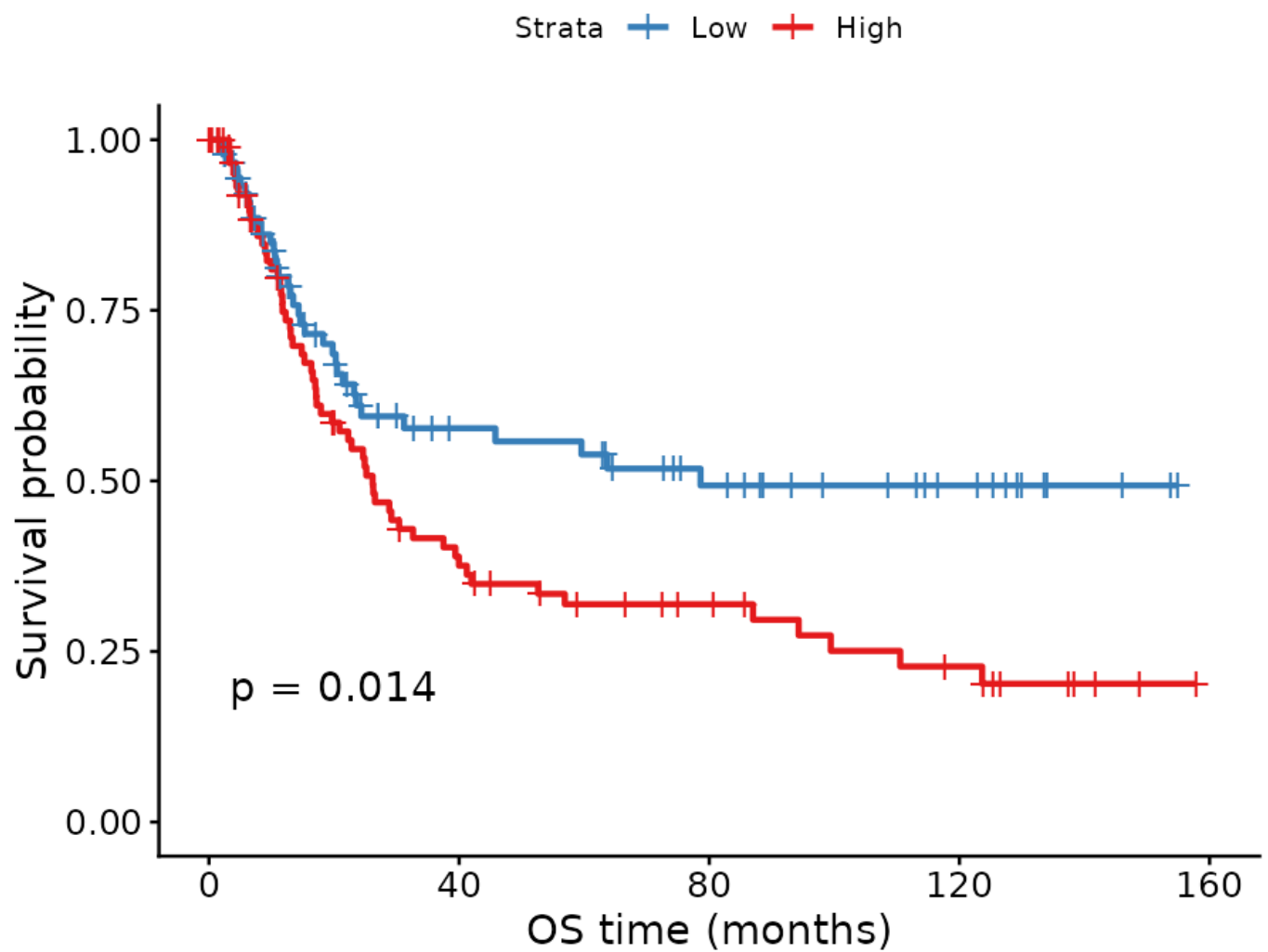


Kaplan-Meier stratification, ACRG/GSE62254 (validation; log-rank $p = 8.6e-5$).

[results/validation/KM_ACRG.png](#)

Figure 23

GSE15459 (validation)

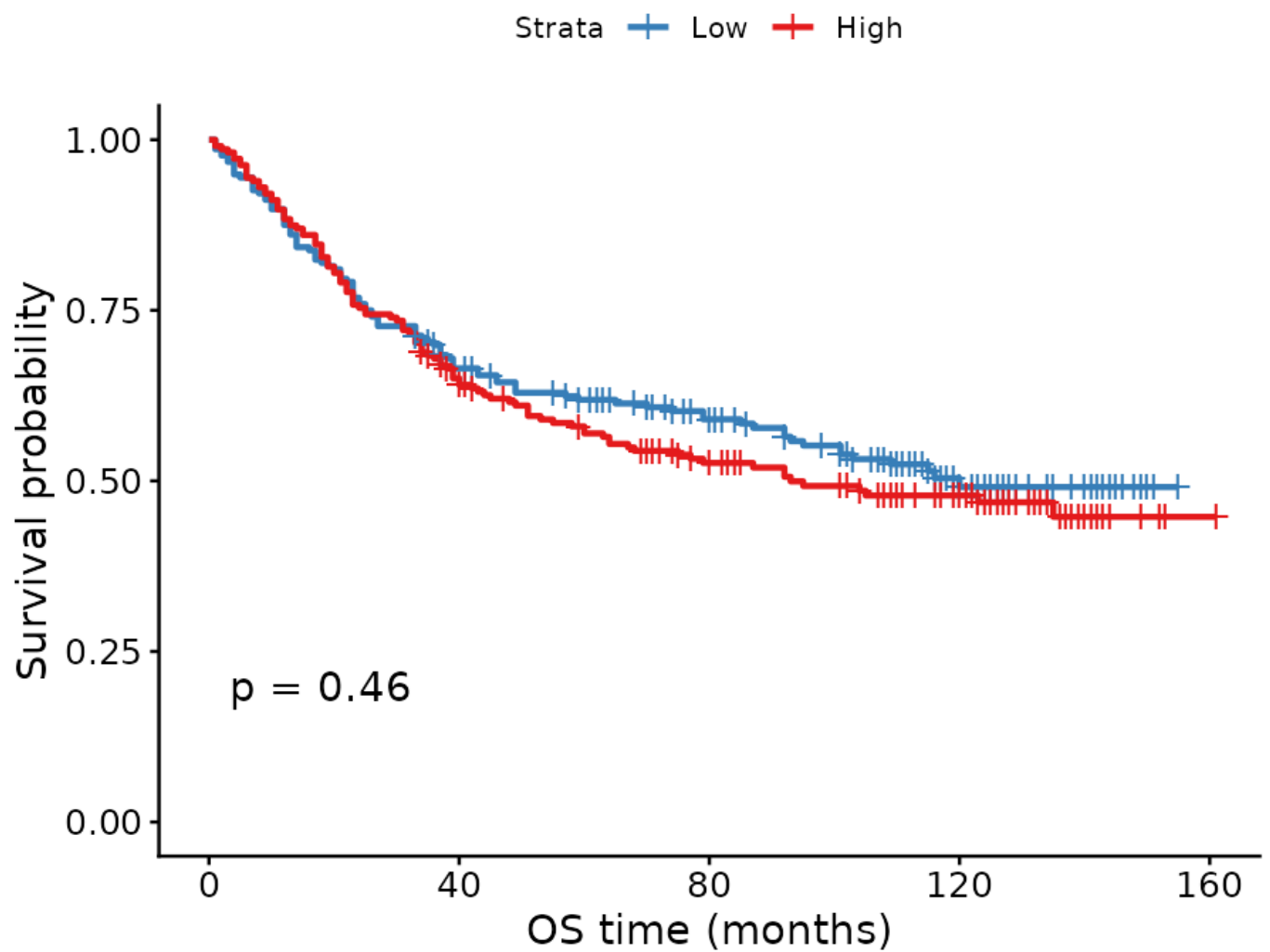


Kaplan-Meier stratification, GSE15459 (validation; log-rank $p = 0.014$).

[results/validation_multi/KM_GSE15459.png](#)

Figure 24

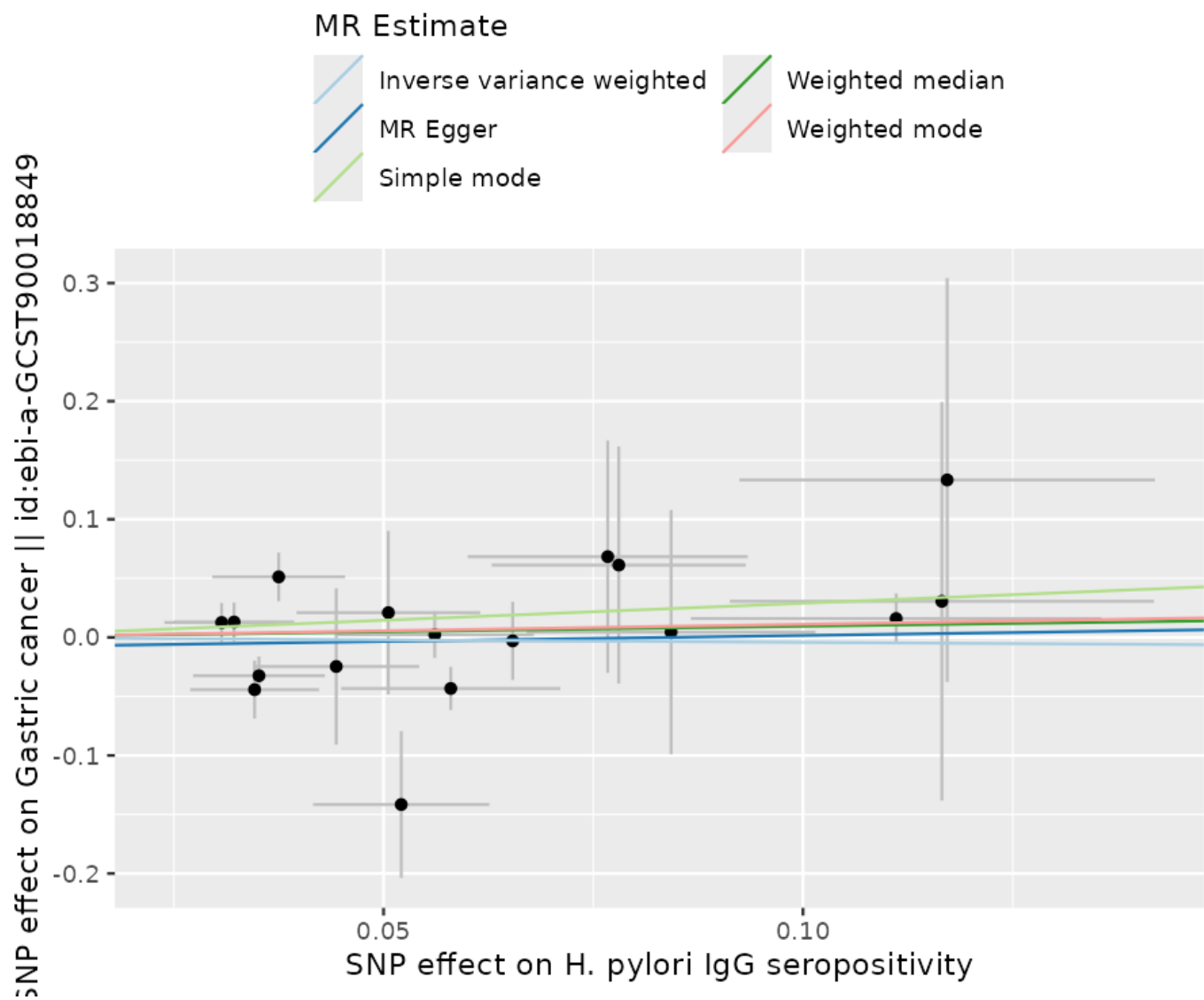
GSE84437 (validation)



Kaplan-Meier stratification, GSE84437 (validation; log-rank $p = 0.46$ - shown transparently as a negative).

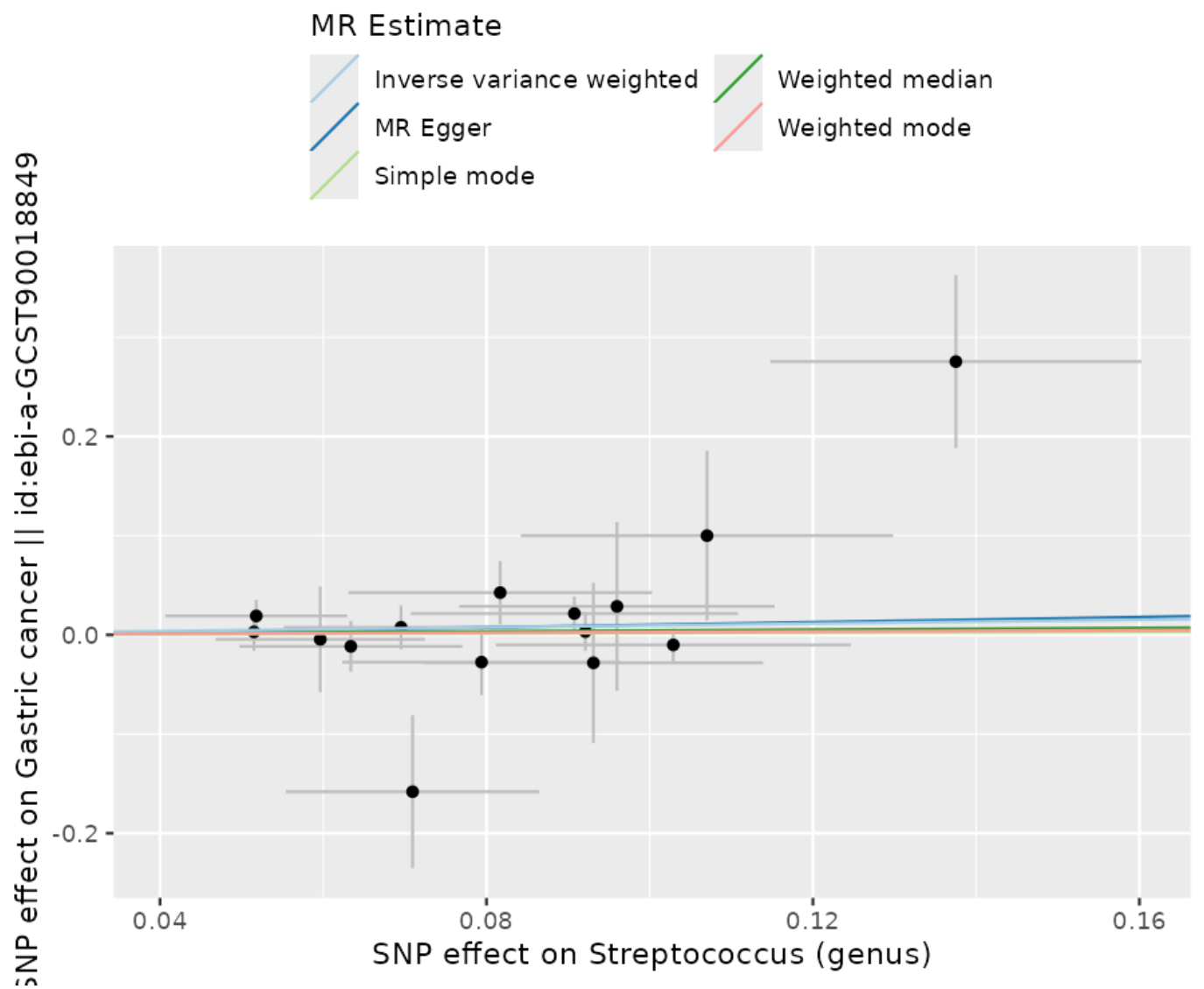
[results/validation_multi/KM_GSE84437.png](#)

Figure 25



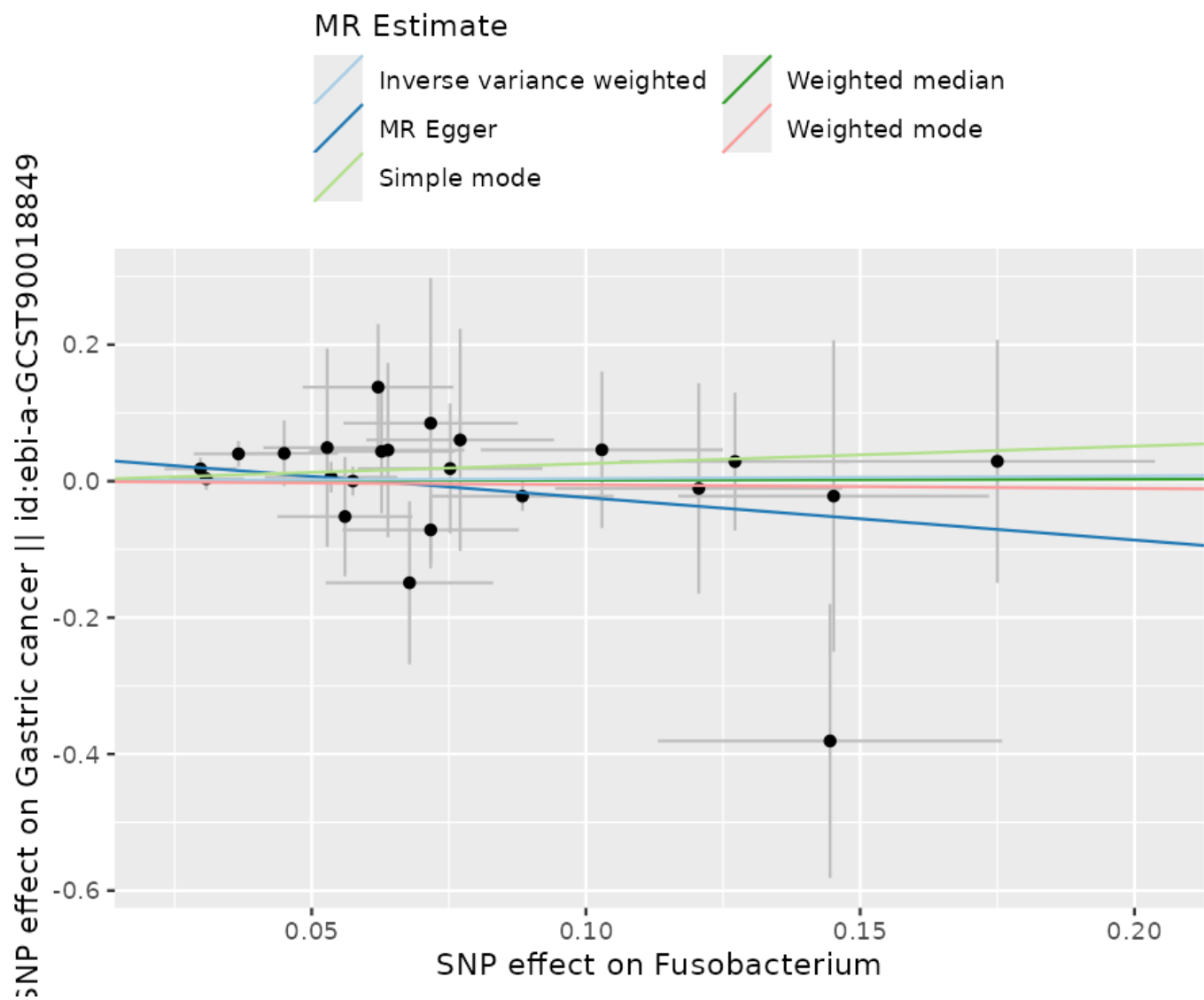
Mendelian-randomisation SNP-effect scatter: anti-H. pylori IgG seropositivity versus gastric cancer (European outcome).
results/mr_real/scatter_H_pylori_IgG_seropositivity.png

Figure 26



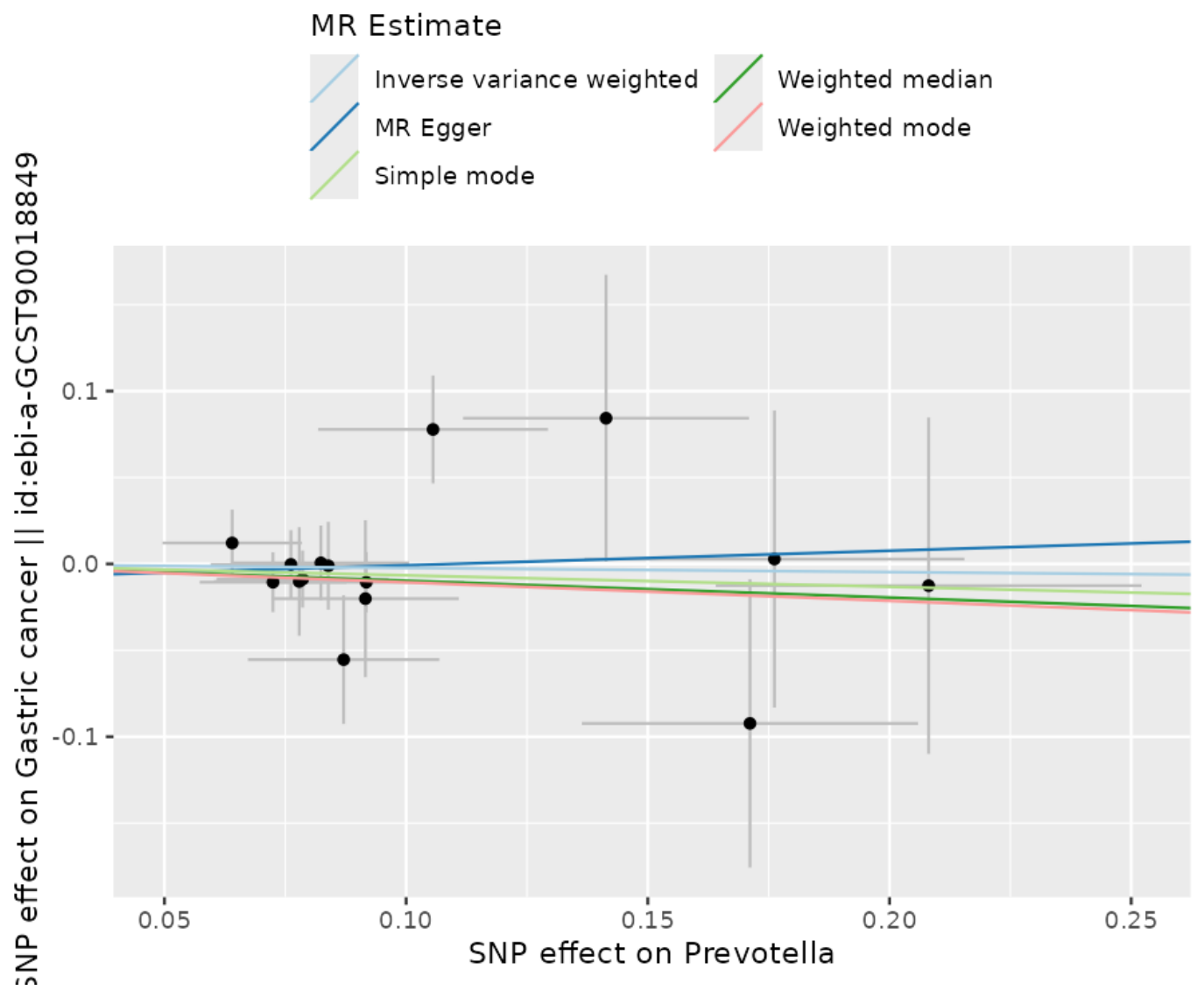
Mendelian-randomisation SNP-effect scatter: Streptococcus (genus) versus gastric cancer (European outcome).
results/mr_real/scatter_Streptococcus_genus_.png

Figure 27



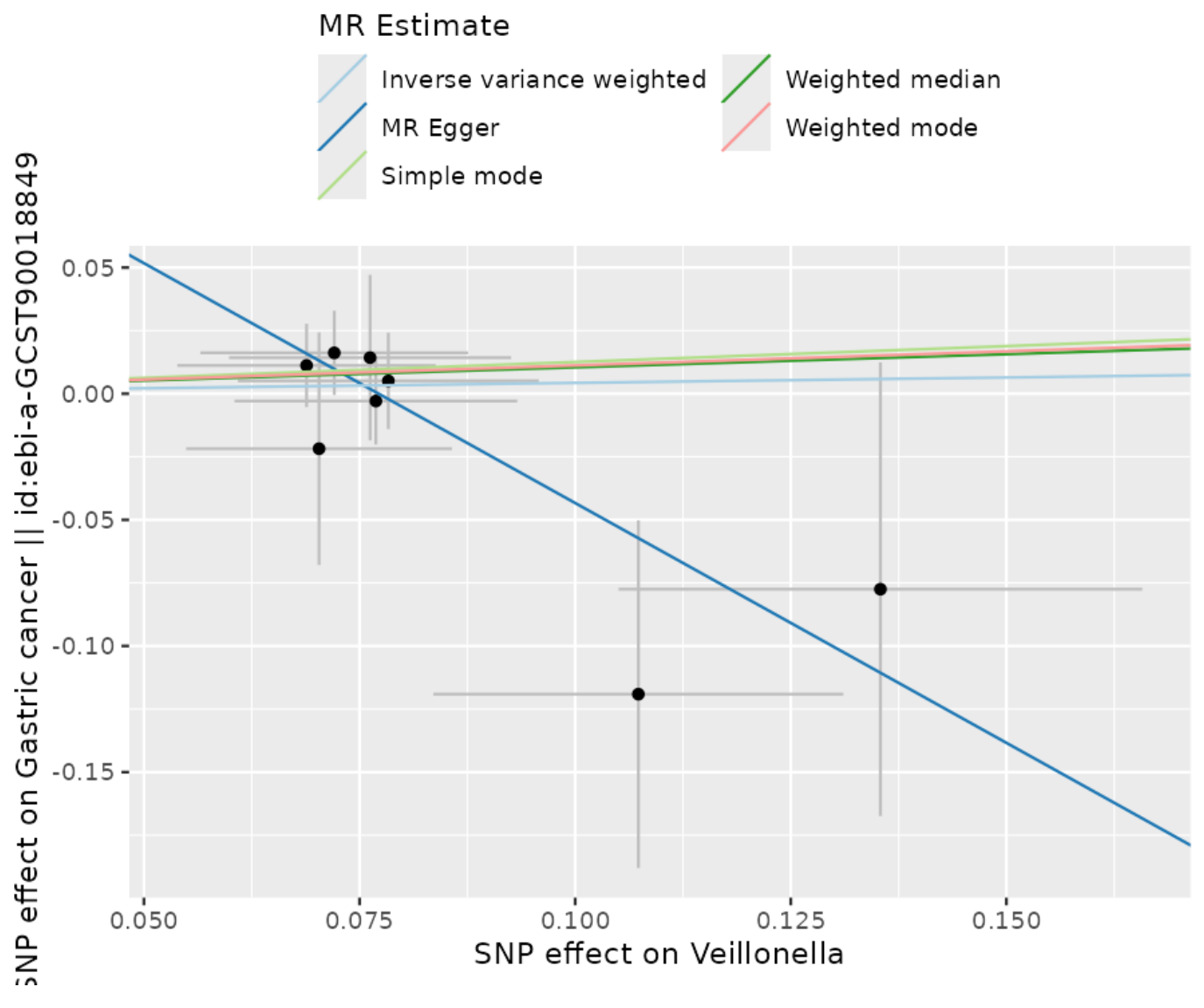
Mendelian-randomisation SNP-effect scatter: Fusobacterium versus gastric cancer (European outcome).
results/mr_real/scatter_Fusobacterium.png

Figure 28



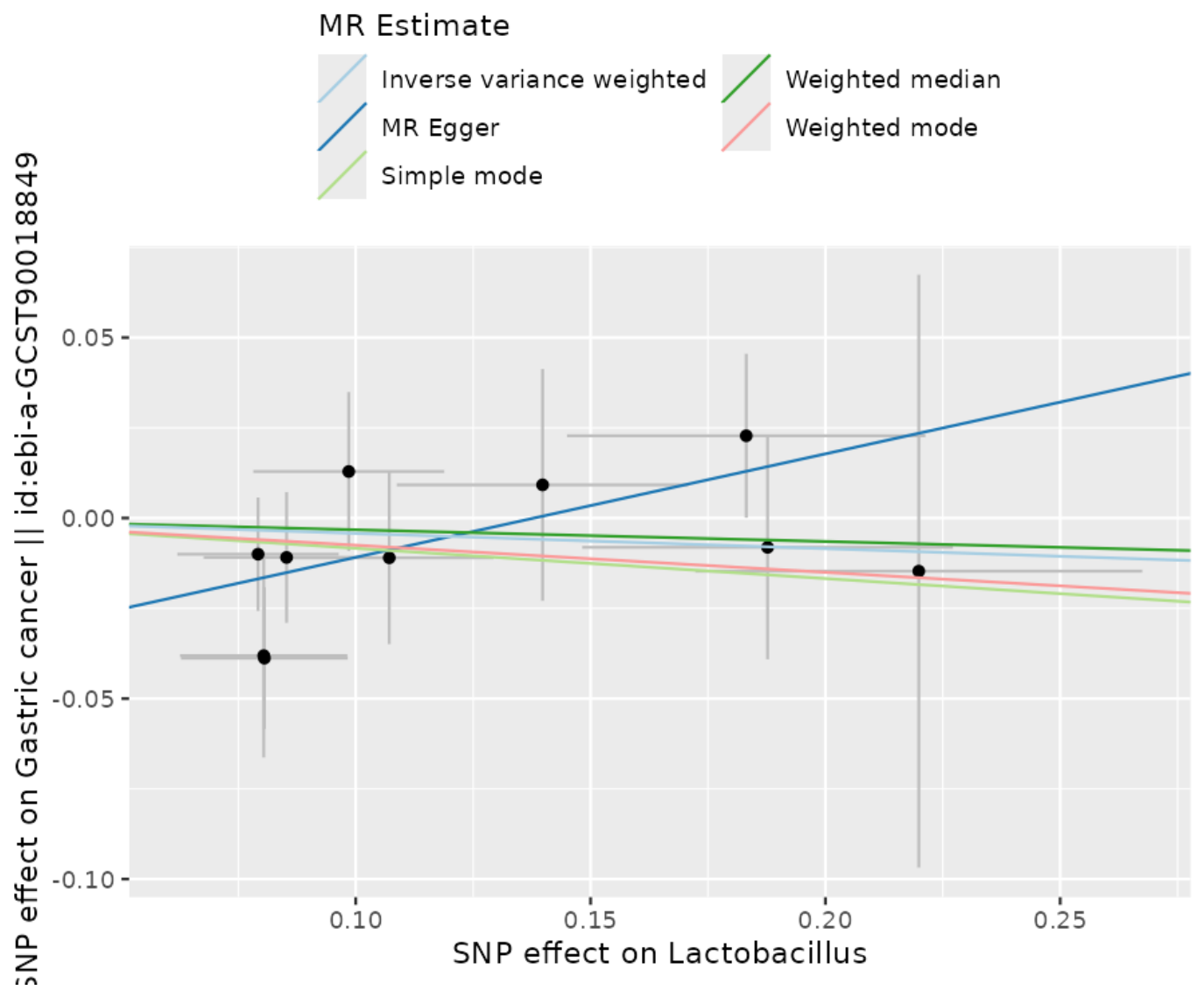
Mendelian-randomisation SNP-effect scatter: Prevotella versus gastric cancer (European outcome).
results/mr_real/scatter_Prevotella.png

Figure 29



Mendelian-randomisation SNP-effect scatter: Veillonella versus gastric cancer (European outcome).
results/mr_real/scatter_Veillonella.png

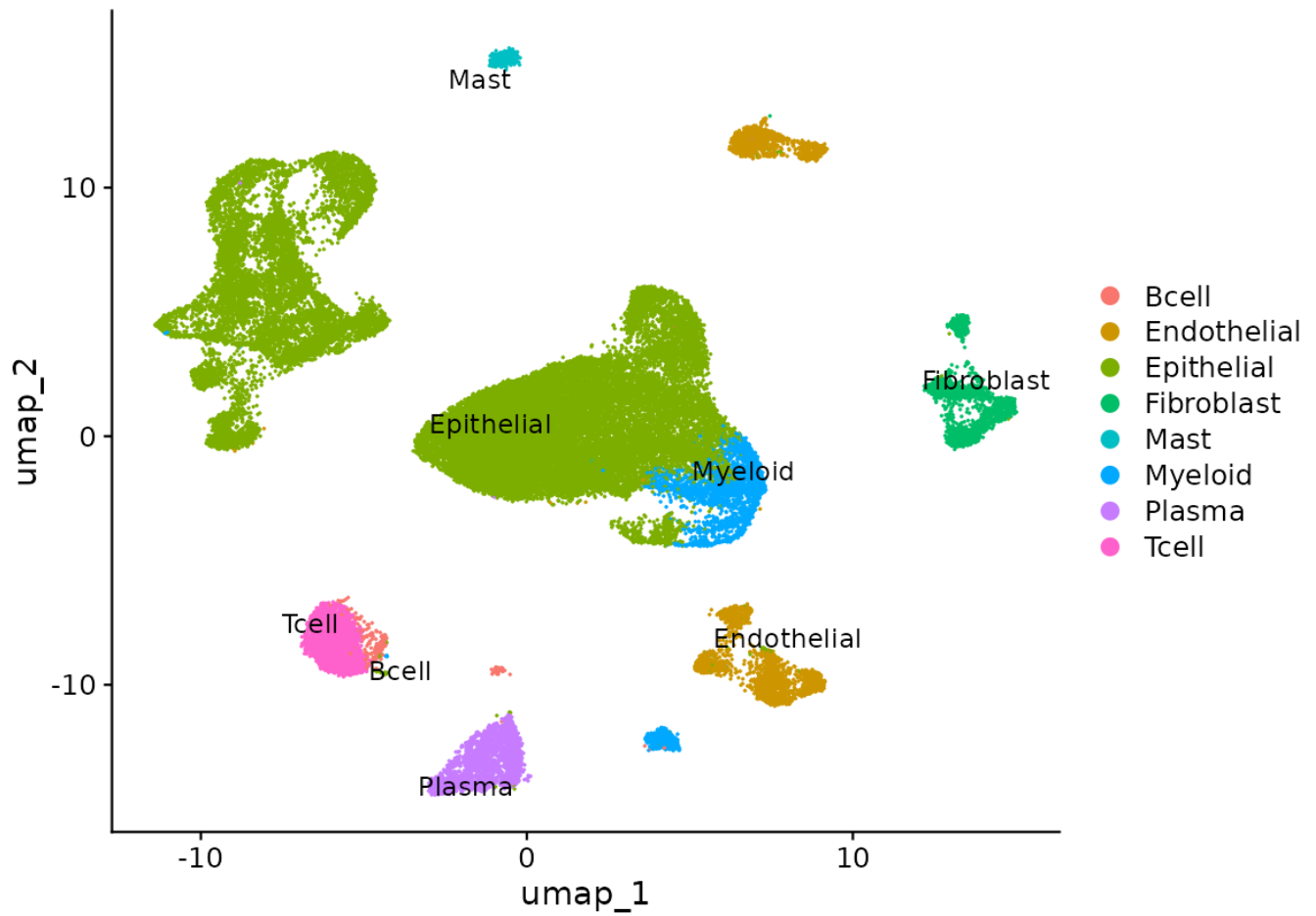
Figure 30



Mendelian-randomisation SNP-effect scatter: Lactobacillus versus gastric cancer (European outcome).
results/mr_real/scatter_Lactobacillus.png

Supplementary Figure S1

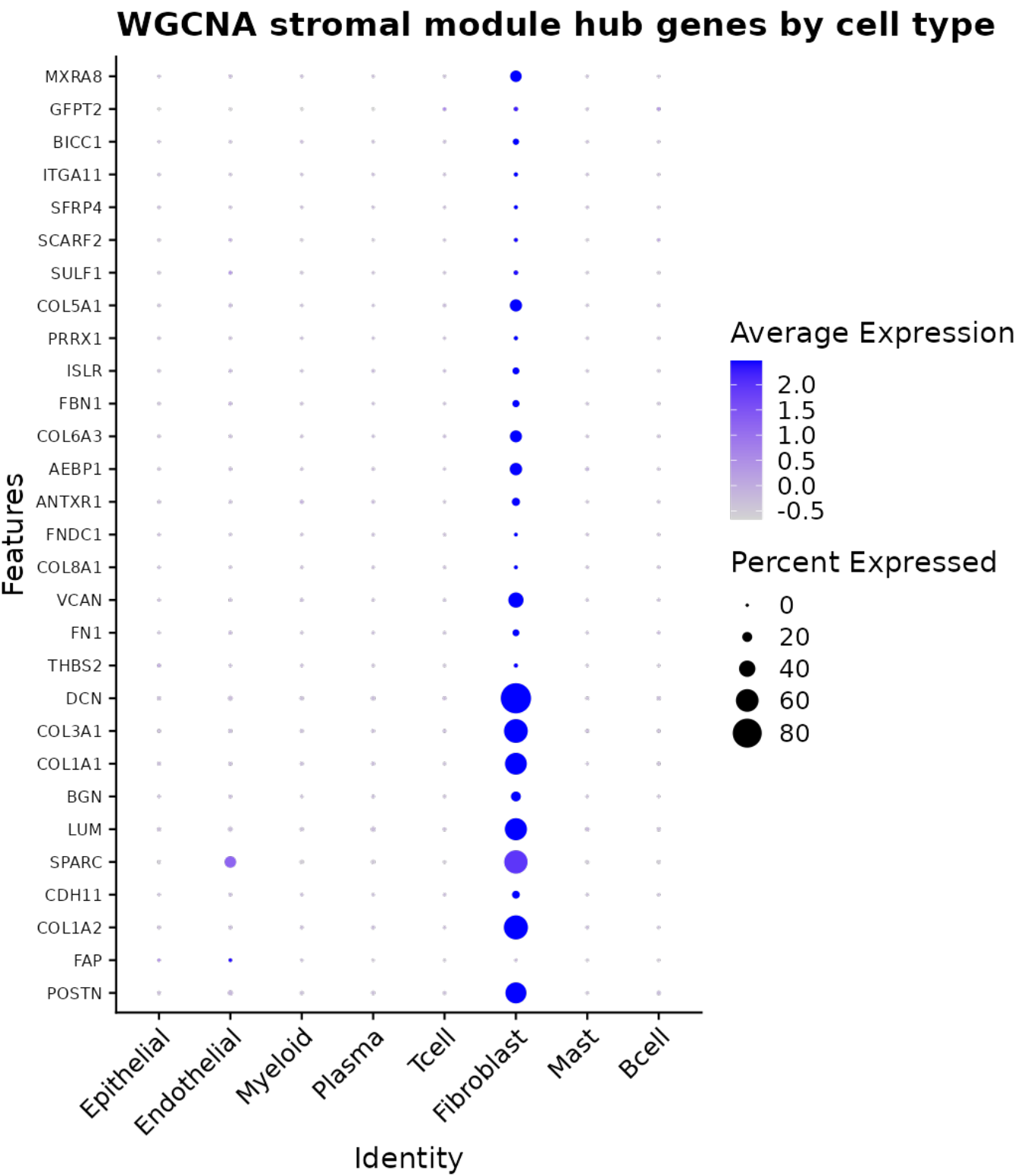
GSE134520 gastric mucosa scRNA-seq — cell types



Single-cell atlas of gastric mucosa (GSE134520): 43,992 cells across eight annotated cell types.

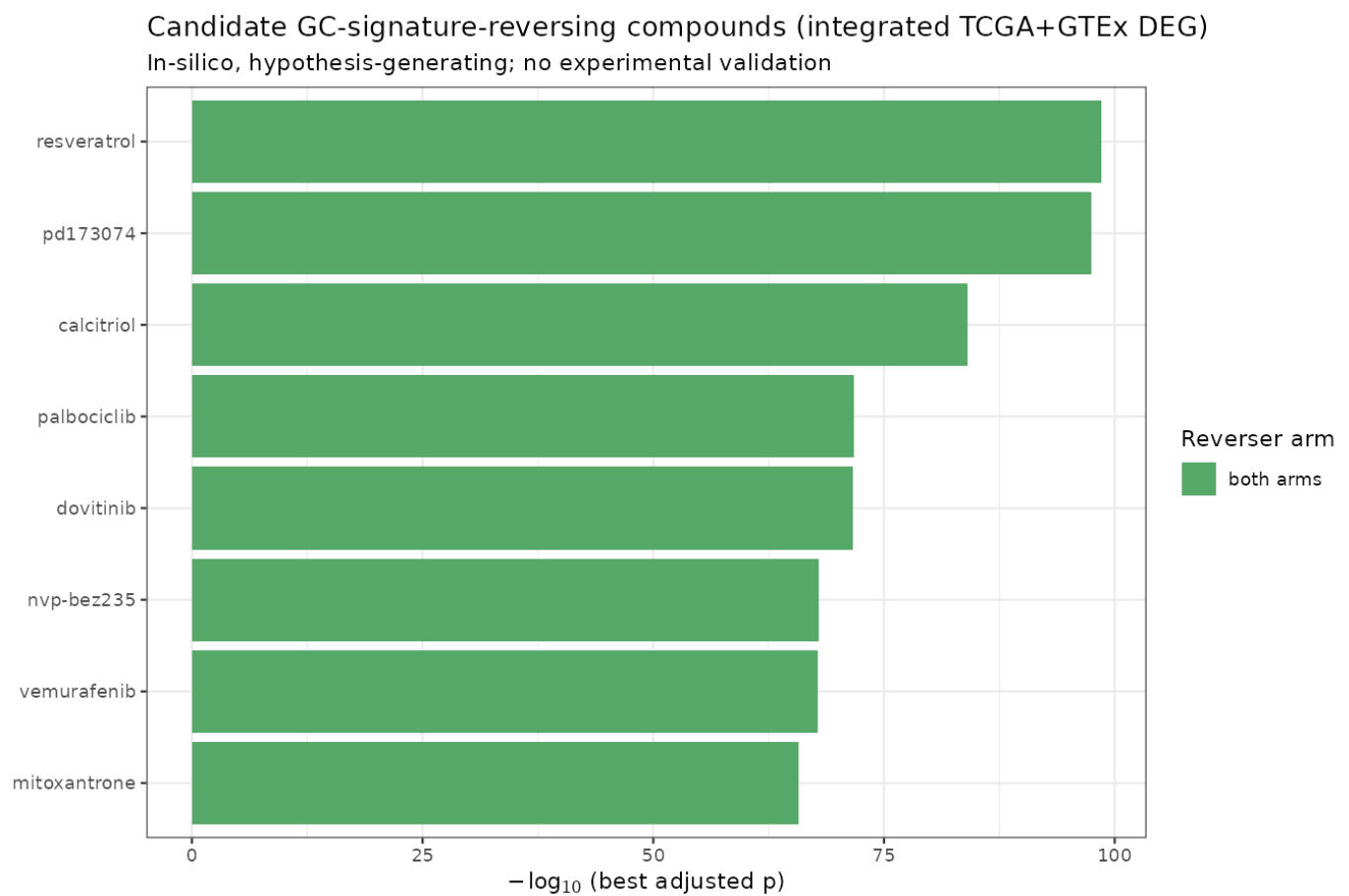
results/scrna/UMAP_celltypes.png

Supplementary Figure S2



Stromal module hub-gene expression by cell type.
results/scrna/DotPlot_stromal_module_hub.png

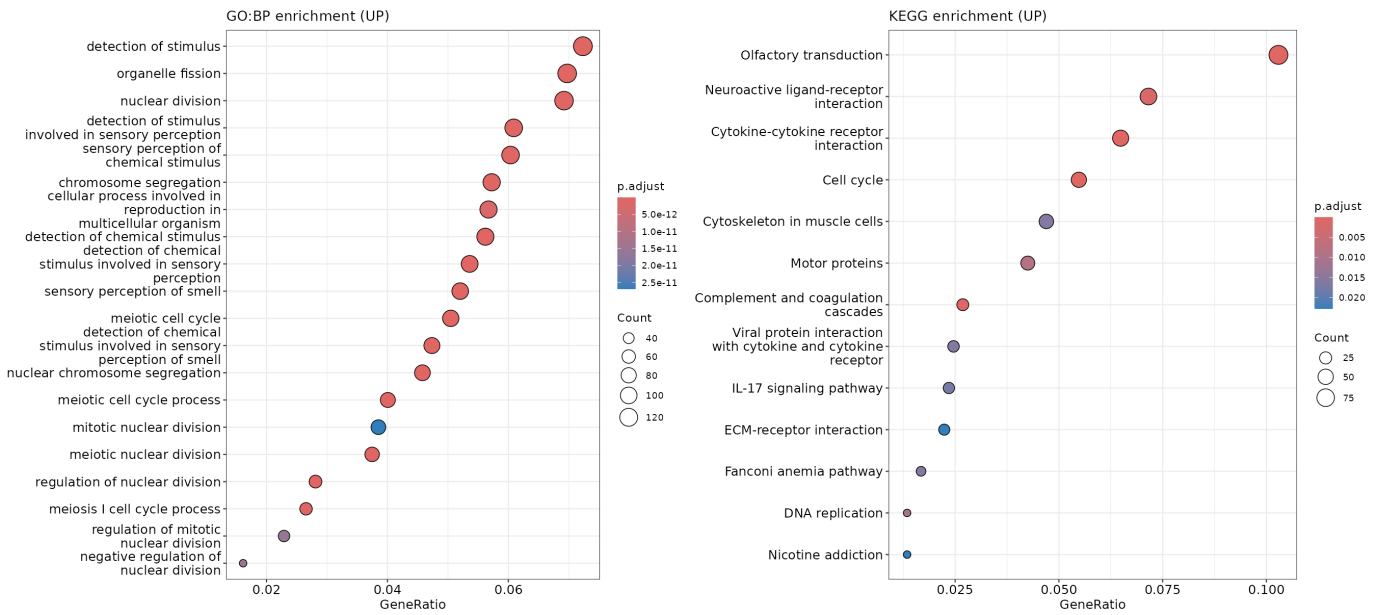
Supplementary Figure S3



In-silico drug-repurposing candidates ranked by connectivity score.

results/drug_repurposing_integrated/top_candidate_drugs.png

Supplementary Figure S4

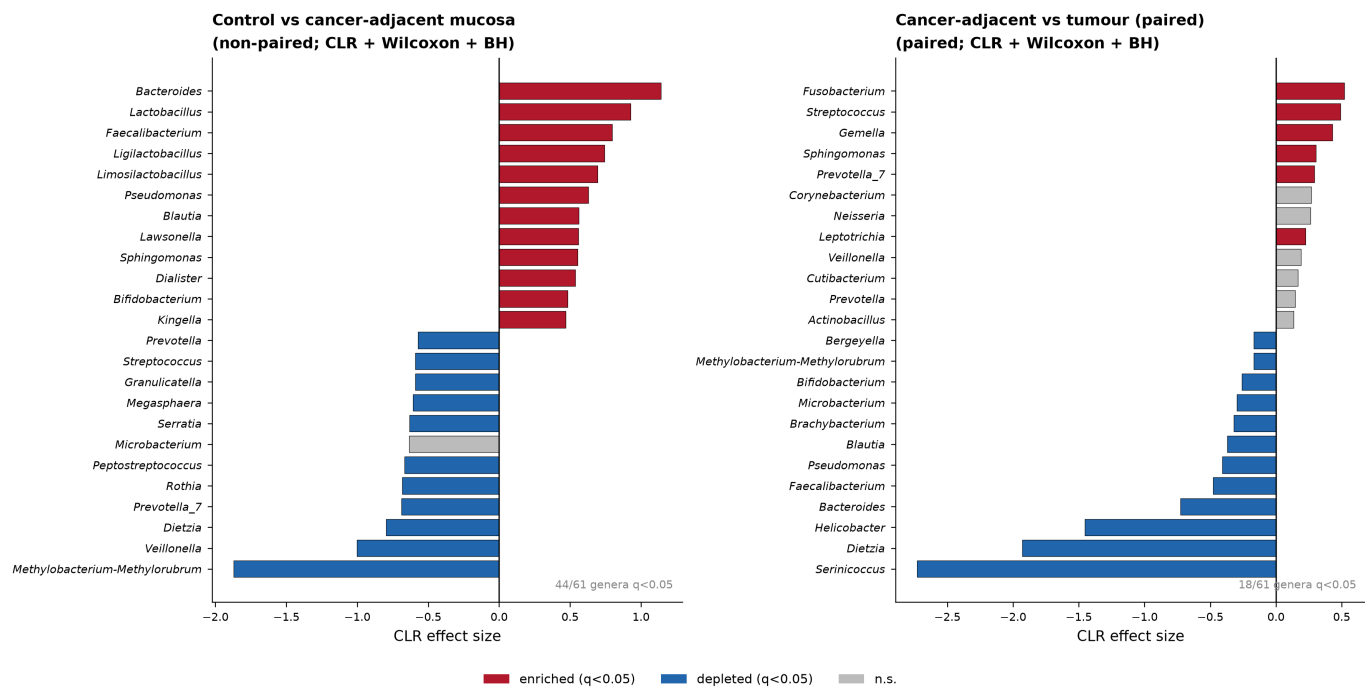


Over-representation analysis (GO biological process and KEGG) of tumour-up genes.

results/enrichment/path_ORA_GO_KEGG.png

Supplementary Figure S5

Tissue-microbiome differential abundance (compositional; batch-cautioned)

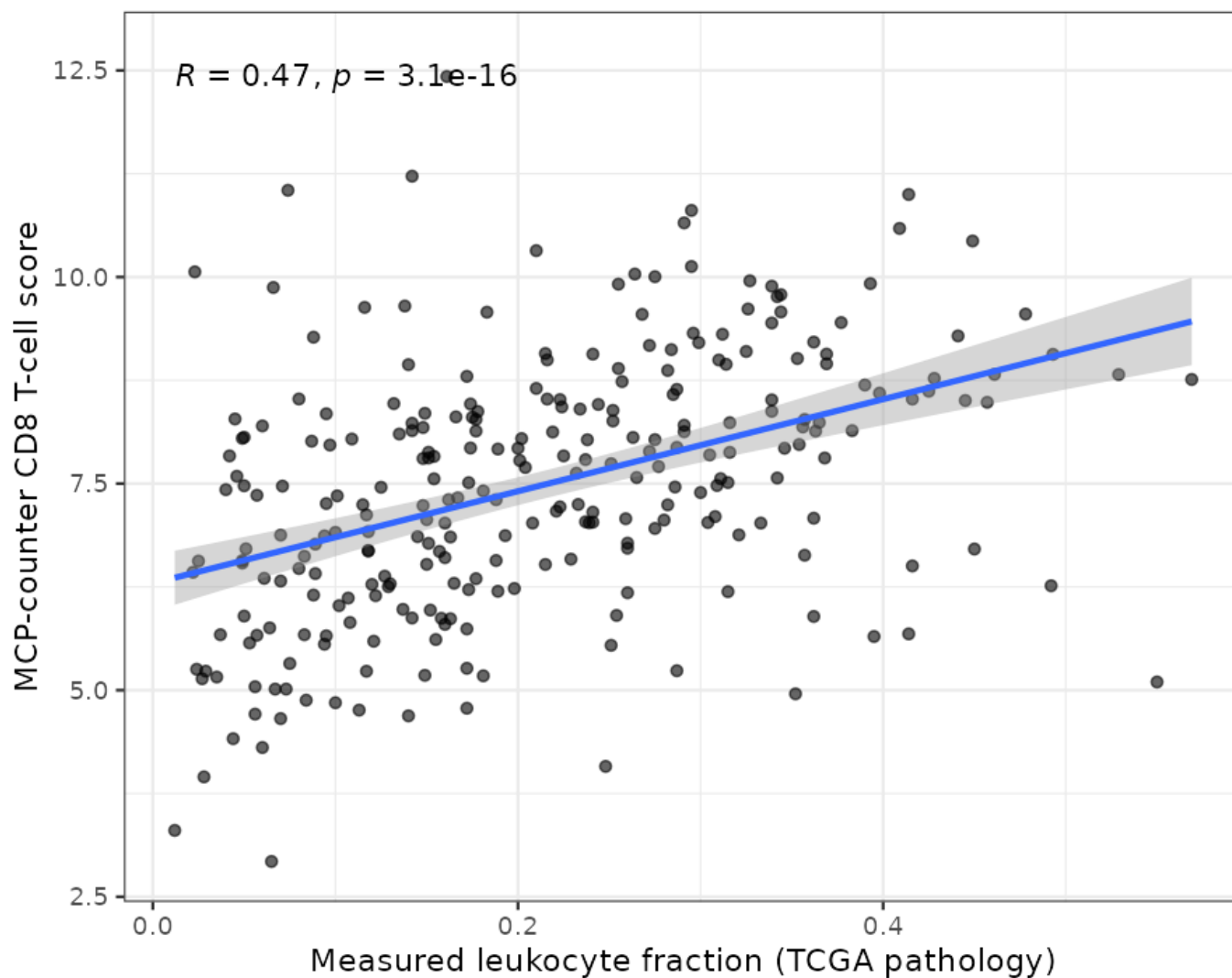


Tissue-microbiome compositional differential abundance (centred log-ratio effect sizes).

results/microbiome_biomarker/da_clr_barplot.png

Supplementary Figure S6

MCP-counter CD8 T cells vs measured leukocyte % (tumors

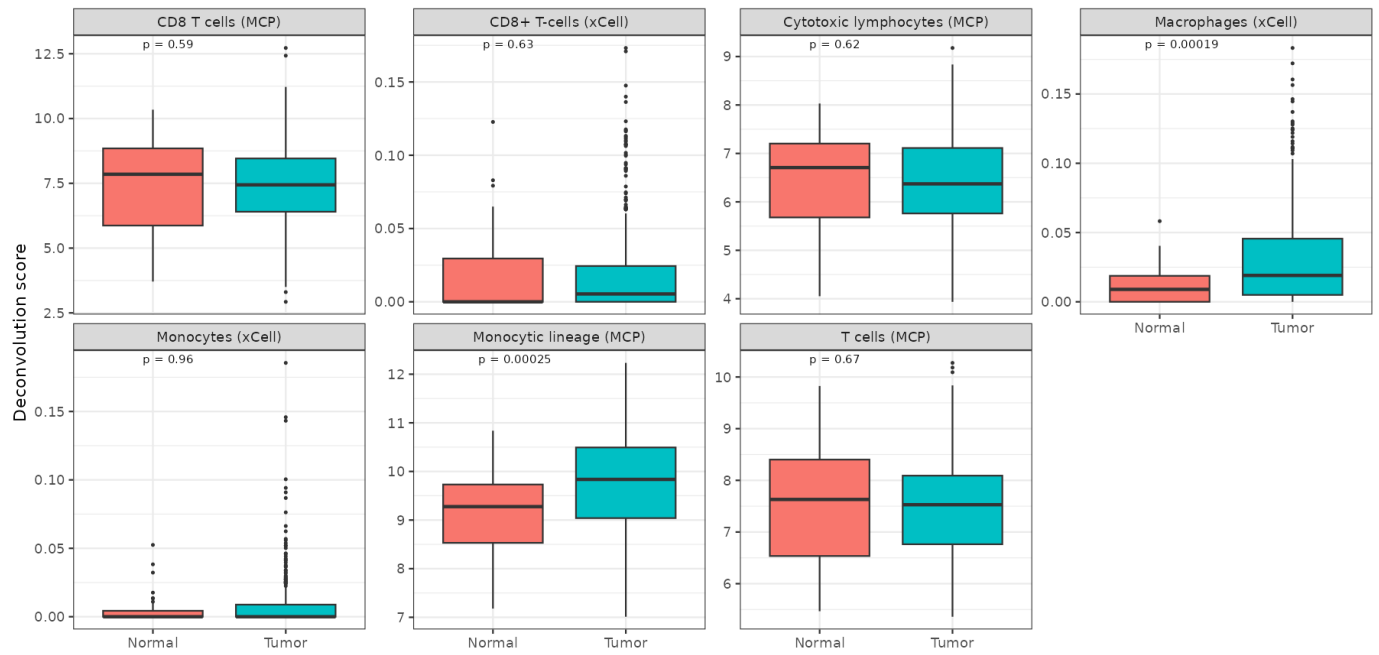


MCP-counter CD8 T-cell score versus measured leukocyte percentage (Spearman rho = 0.468, $p = 3.1e-16$, $n = 272$). The total T-cell score correlates more strongly (rho = 0.666). Neither correlates with measured lymphocyte-infiltration percentage (CD8 rho = 0.020, $p = 0.74$).

results/plots/Immune_validation_scatter.png

Supplementary Figure S7

Immune populations: Tumor vs Normal (TCGA-STAD)

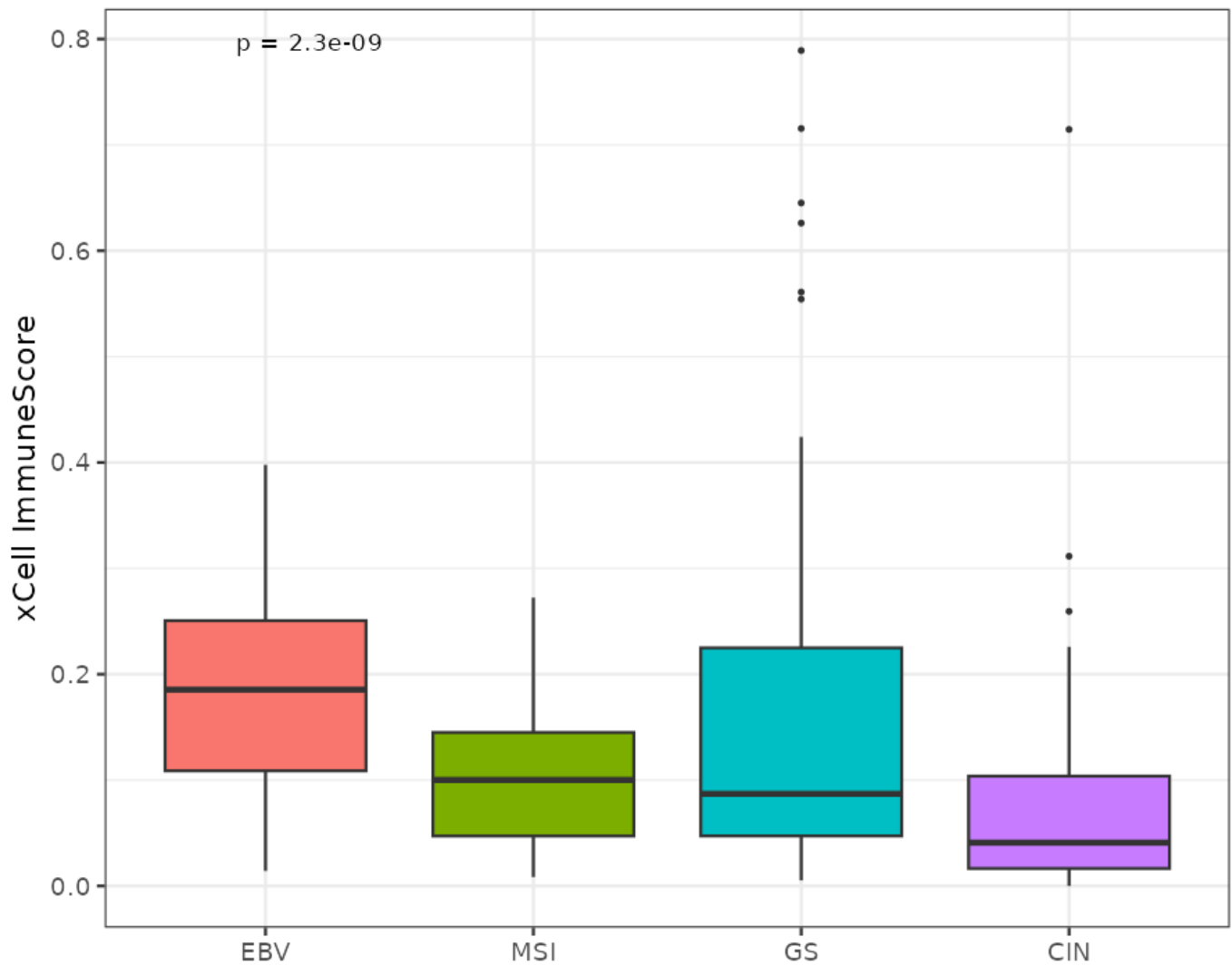


Immune cell-type estimates, tumour versus normal tissue.

results/plots/Immune_tumor_vs_normal.png

Supplementary Figure S8

xCell ImmuneScore by molecular subtype (TCGA-STAD)

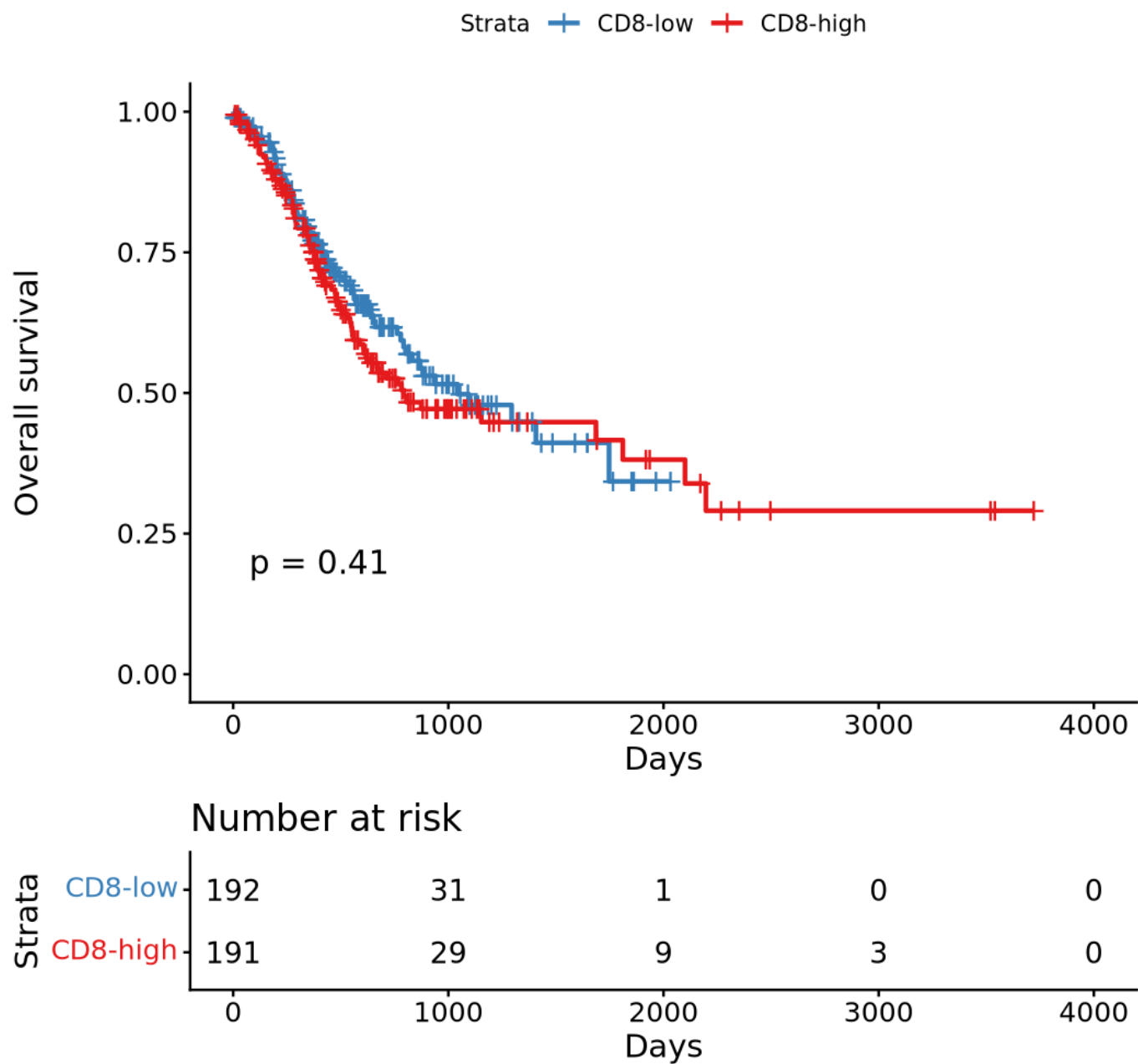


Immune infiltration by Lauren subtype.

results/plots/Immune_by_subtype.png

Supplementary Figure S9

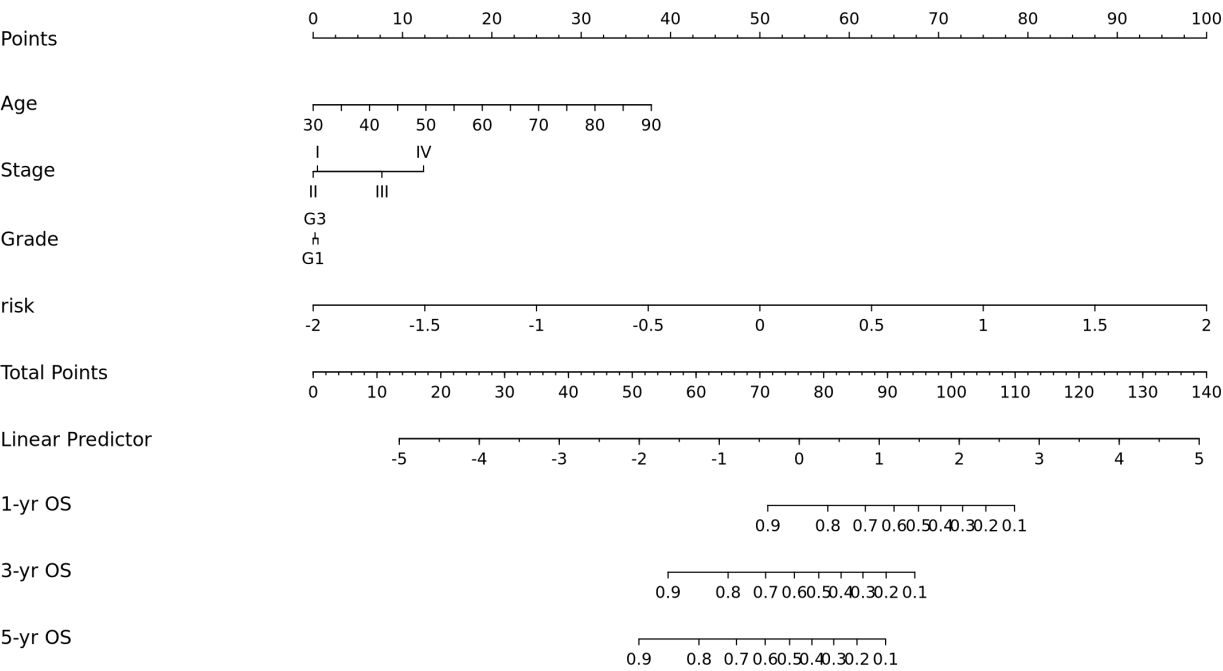
OS by MCP-counter CD8 T-cell score (median split)



CD8 T-cell infiltration and overall survival (HR 1.04, $p = 0.41$; not prognostic).

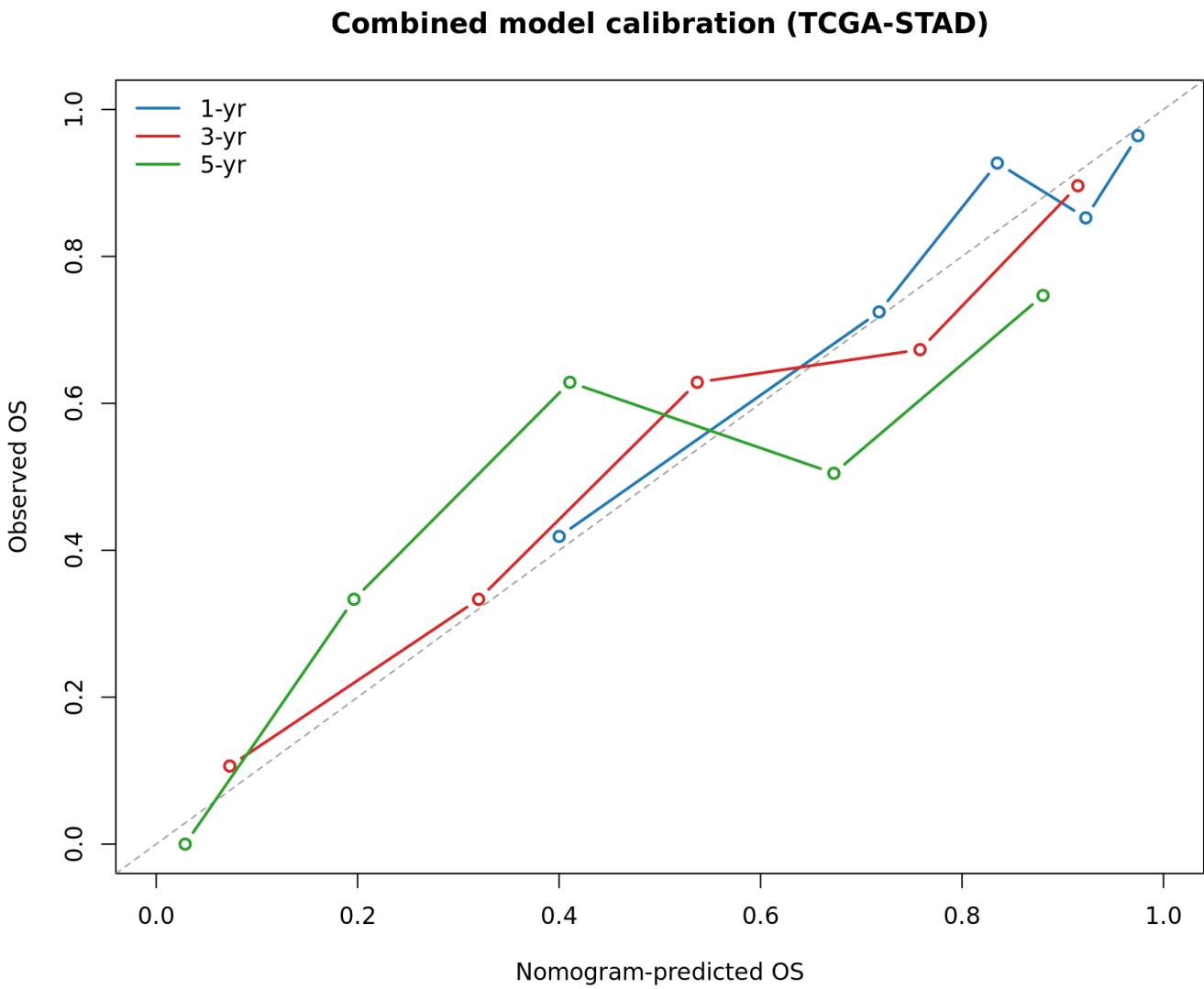
results/plots/Immune_CD8_survival_KM.png

Supplementary Figure S10



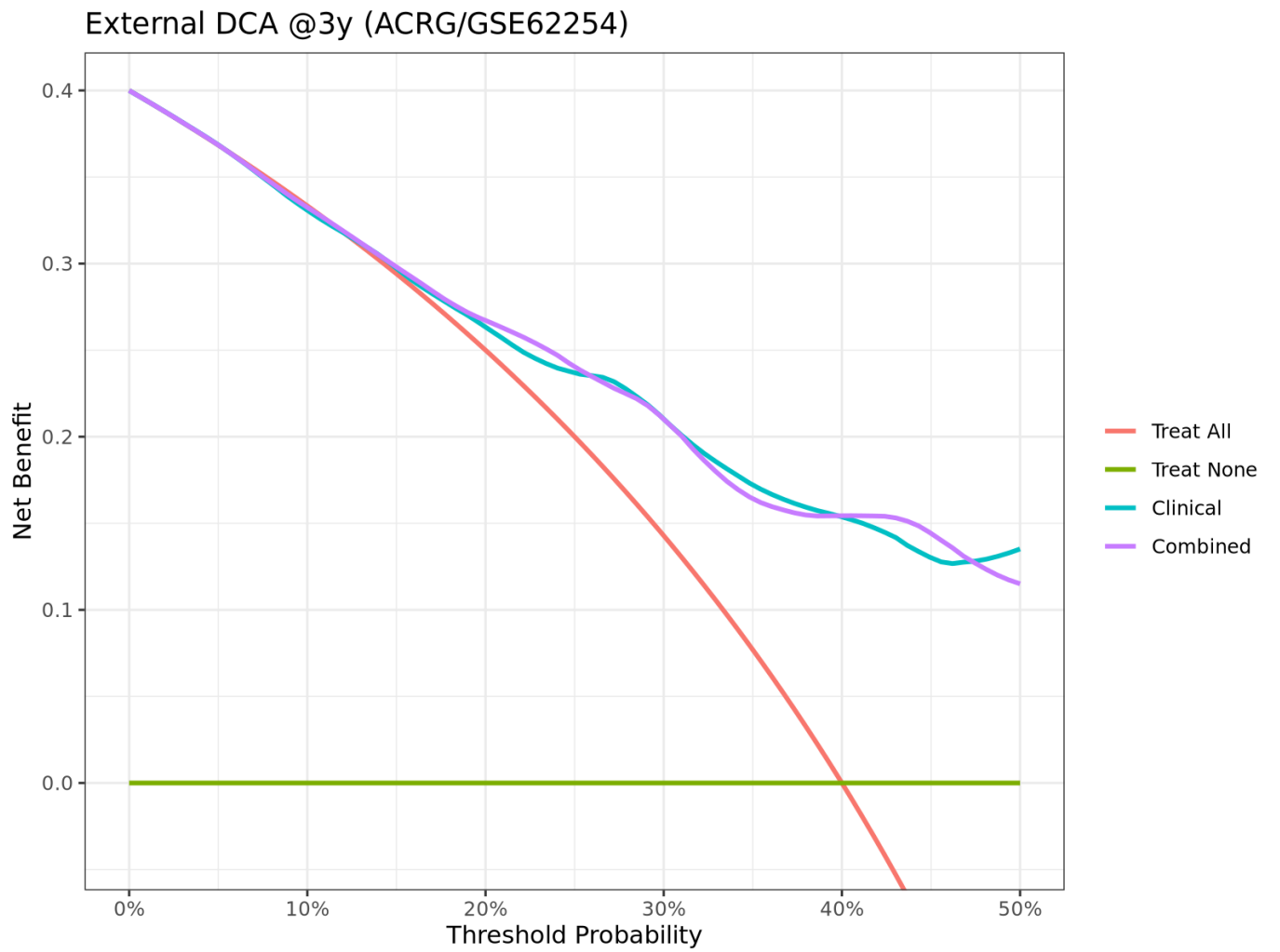
Clinical nomogram combining stage, age and the 25-gene signature.

[results/nomogram_combined/combined_nomogram.png](#)



Calibration of the combined model against observed survival.
results/nomogram_combined/calibration_combined.png

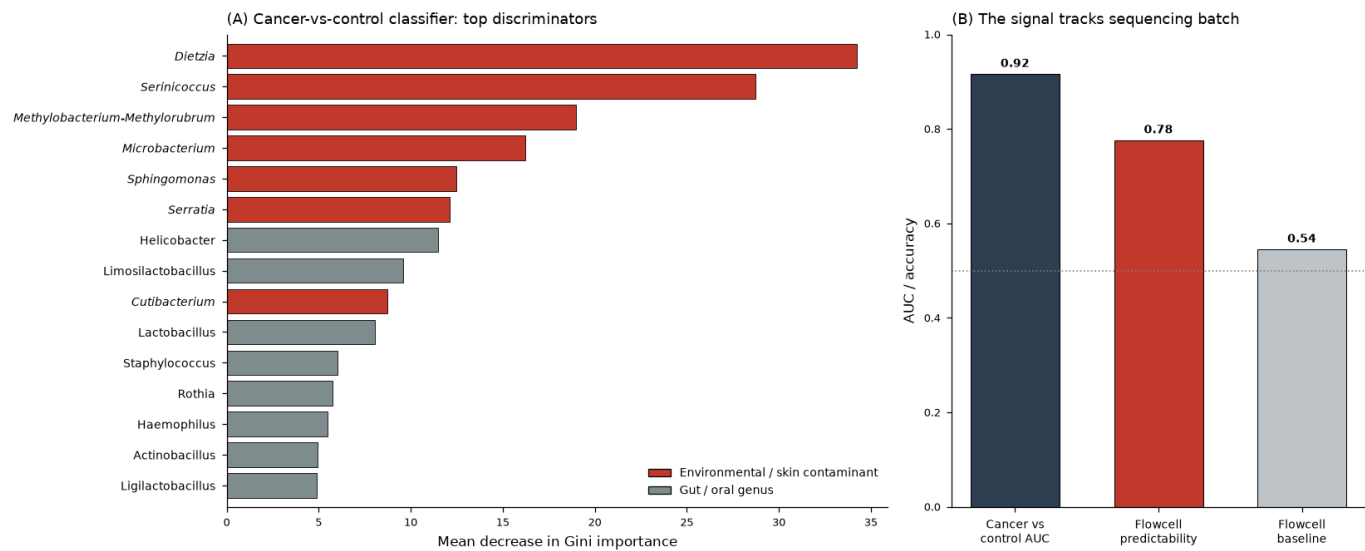
Supplementary Figure S12



External decision-curve analysis in ACRG: no incremental net benefit over standard staging (delta-C +0.005).
results/external_utility_ACRG/DCA_external.png

Supplementary Figure S13

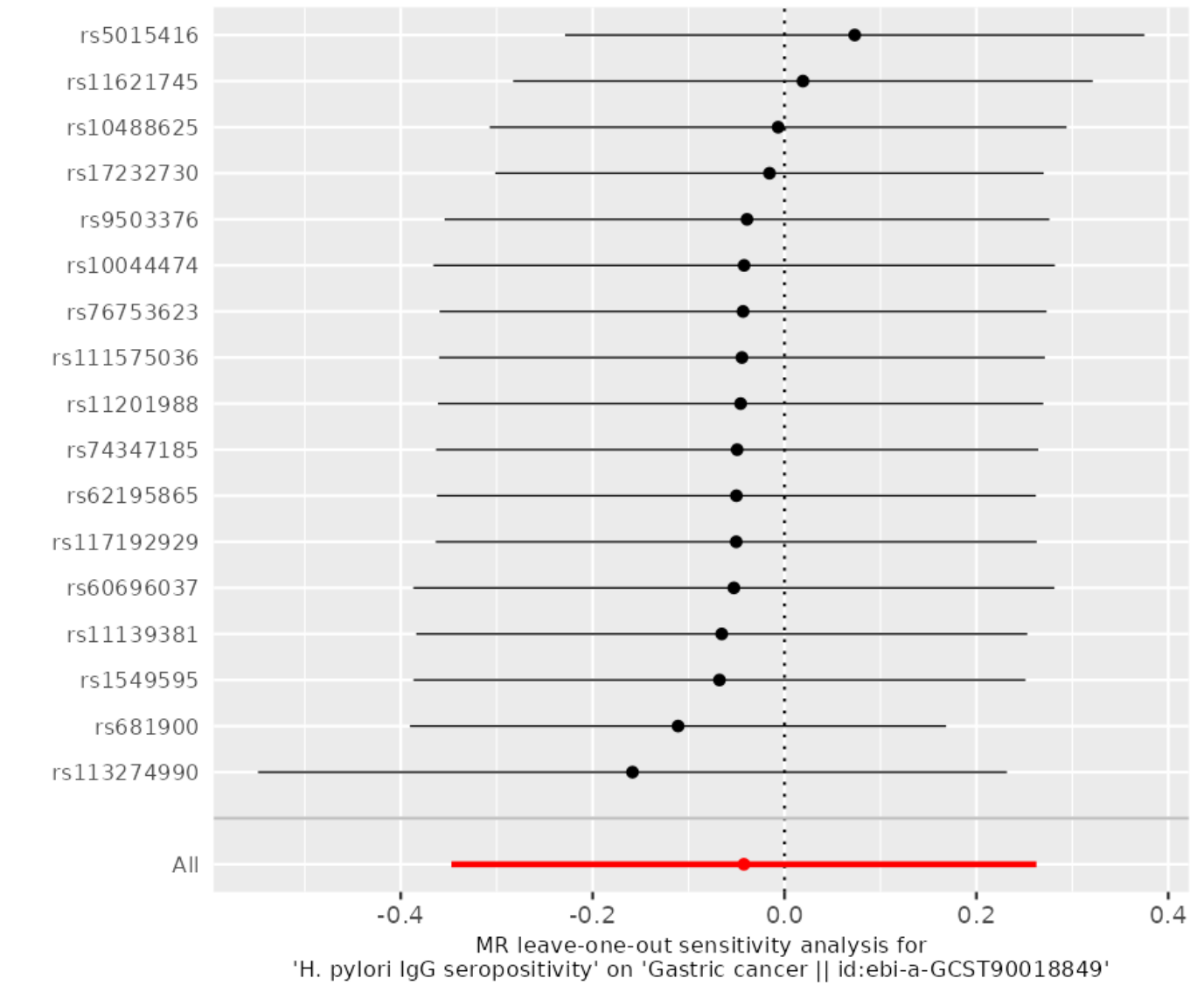
Tumour-microbiome random forest: apparent accuracy is a batch artefact



The tumour-microbiome classifier is a batch artefact: apparent cancer-vs-control AUC 0.916, but flowcell is predictable at 77.6% versus a 54.5% baseline, and top discriminating taxa are environmental contaminants.

results/microbiome_biomarker/rf_batch_classifier.png

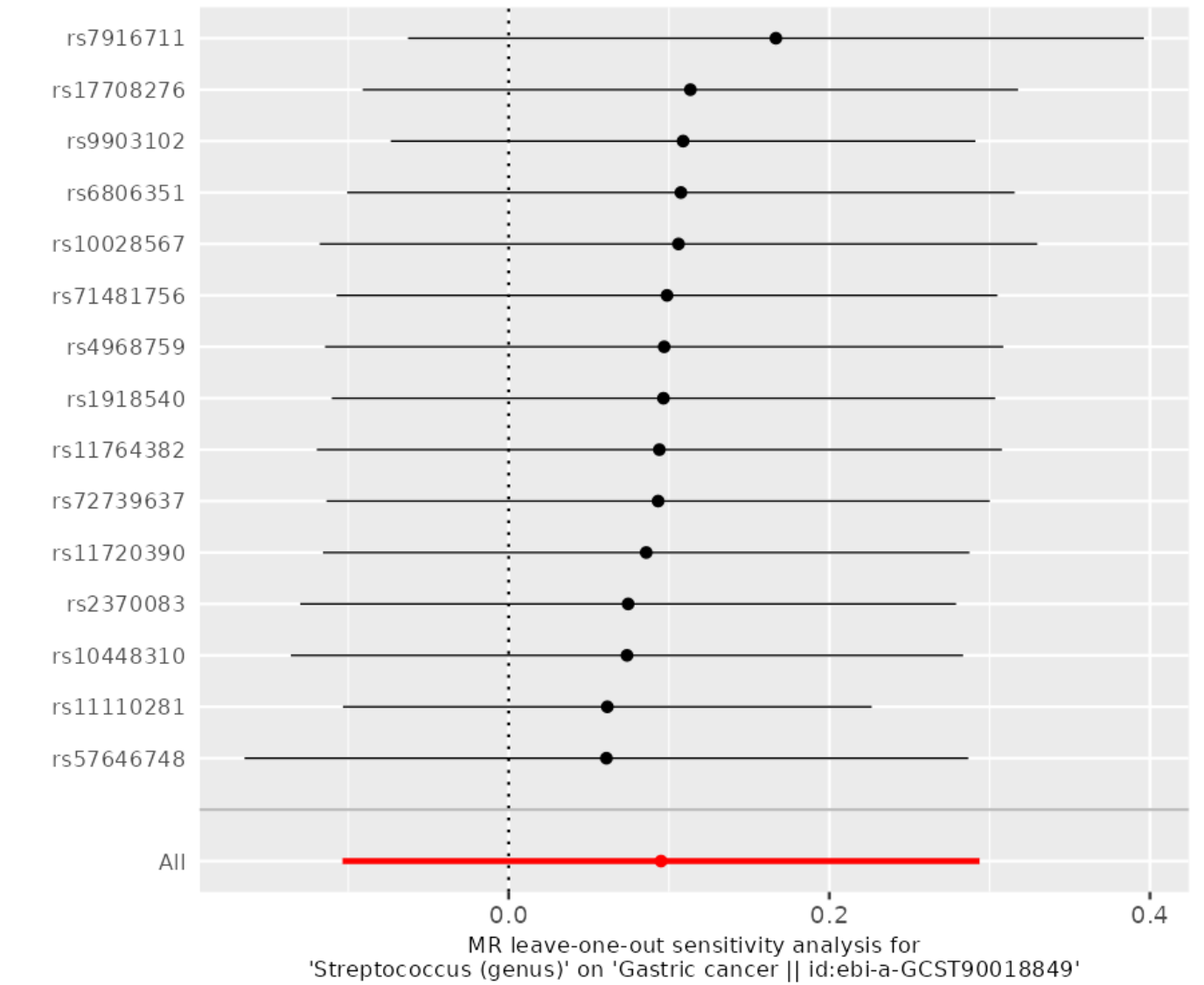
Supplementary Figure S14



Mendelian-randomisation leave-one-out analysis: anti-H. pylori IgG seropositivity.

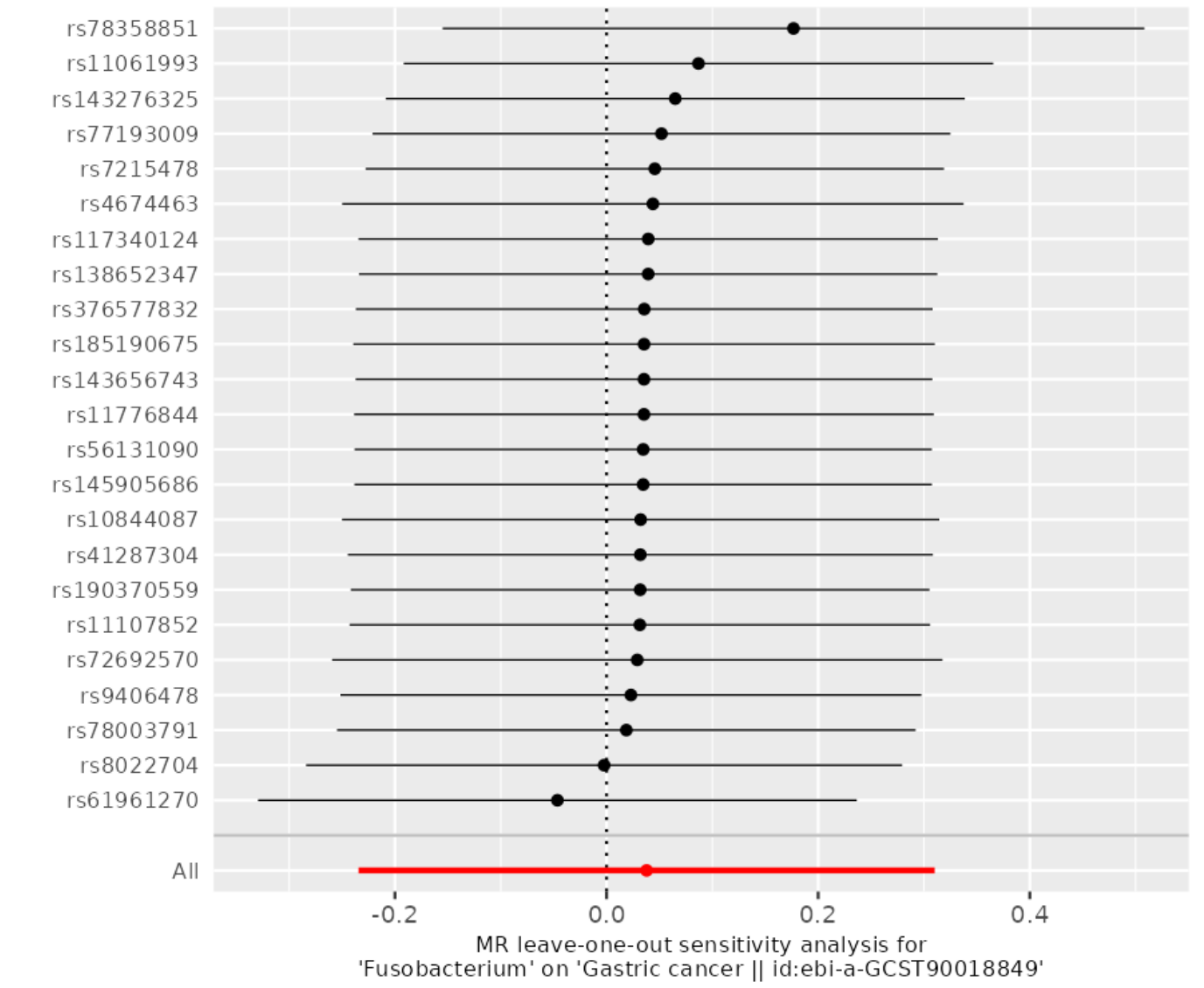
results/mr_real/loo_H_pylori_IgG_seropositivity.png

Supplementary Figure S15



Mendelian-randomisation leave-one-out analysis: Streptococcus (genus).
results/mr_real/loo_Streptococcus__genus_.png

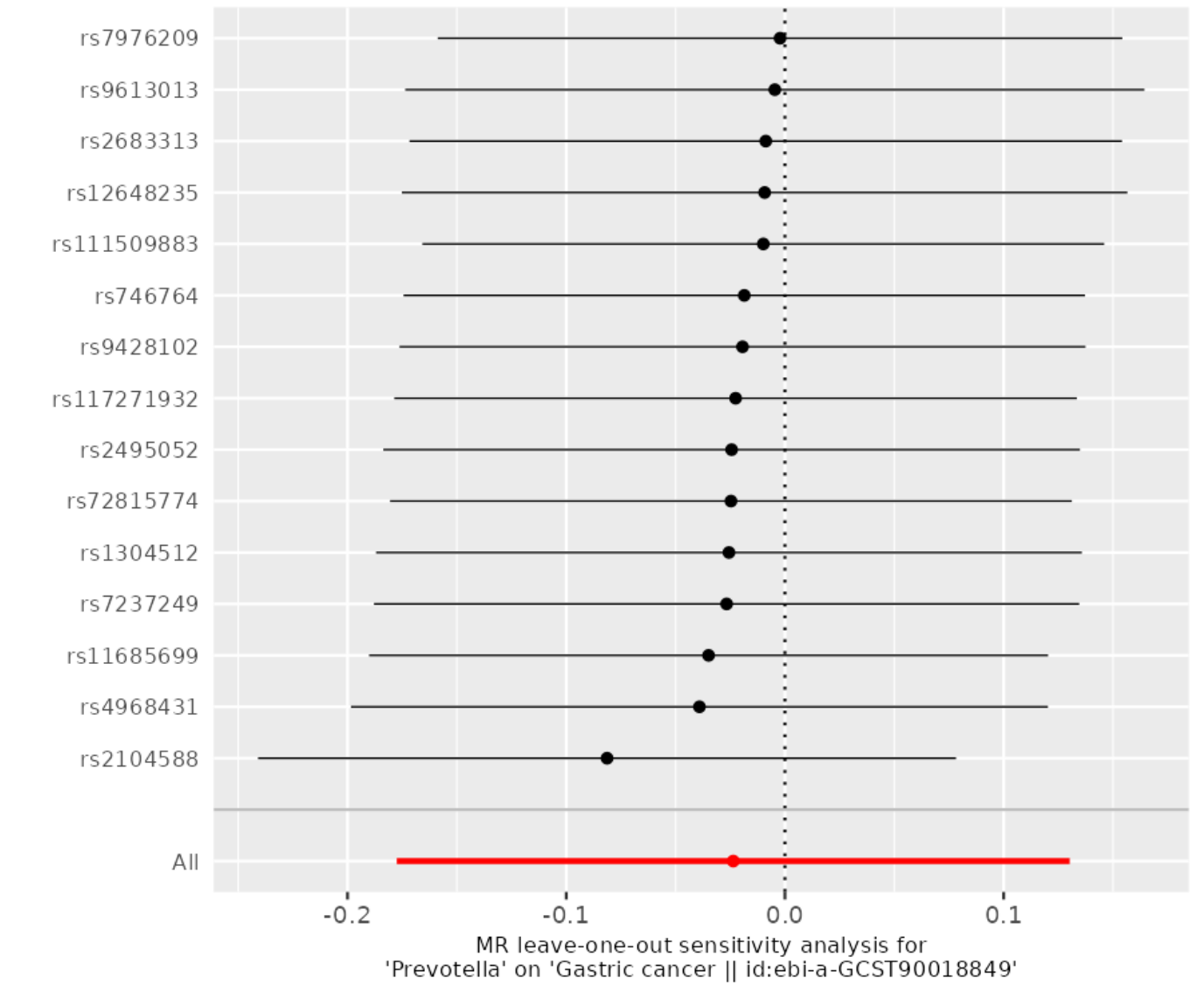
Supplementary Figure S16



Mendelian-randomisation leave-one-out analysis: Fusobacterium.

results/mr_real/loo_Fusobacterium.png

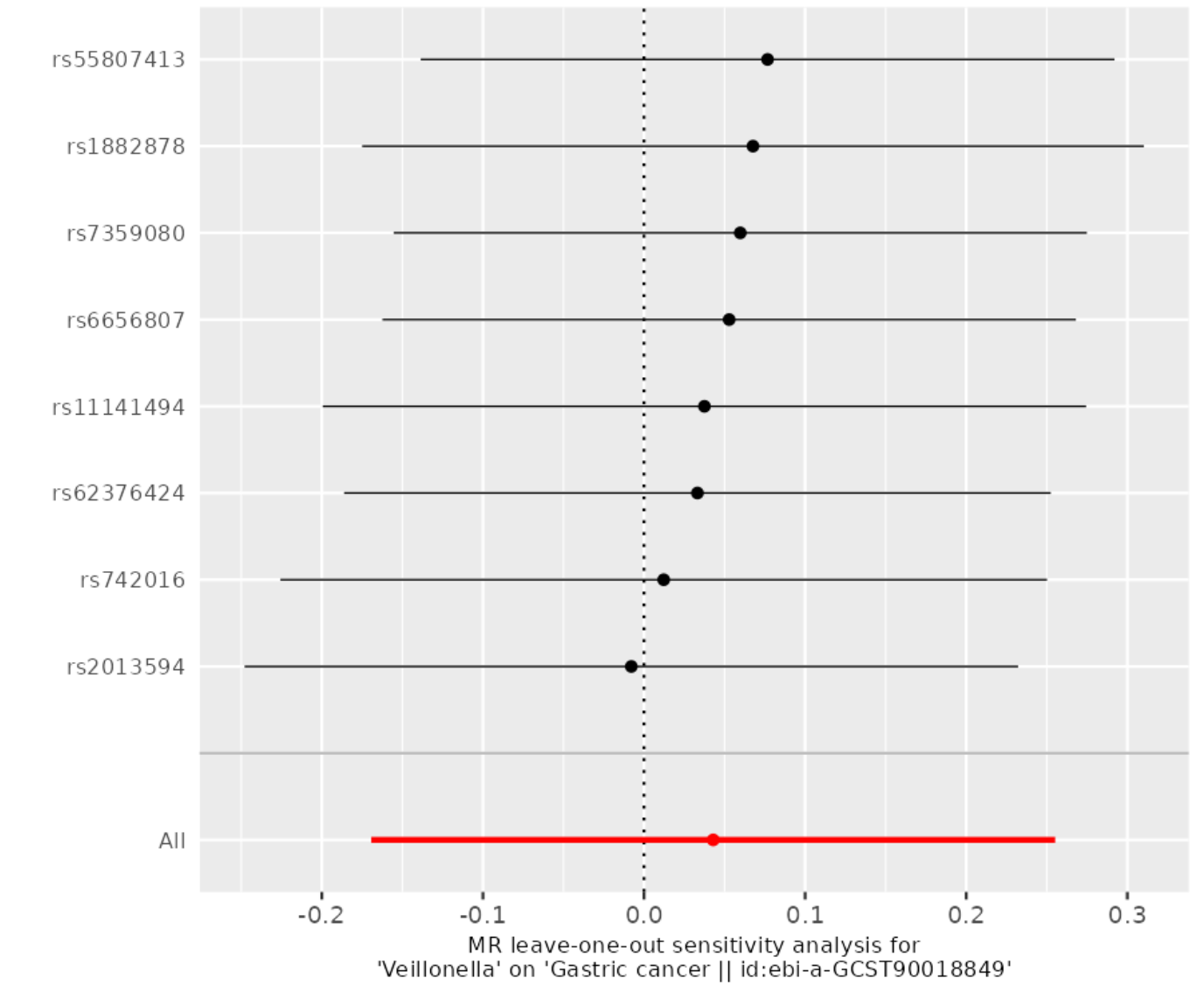
Supplementary Figure S17



Mendelian-randomisation leave-one-out analysis: Prevotella.

results/mr_real/loo_Prevotella.png

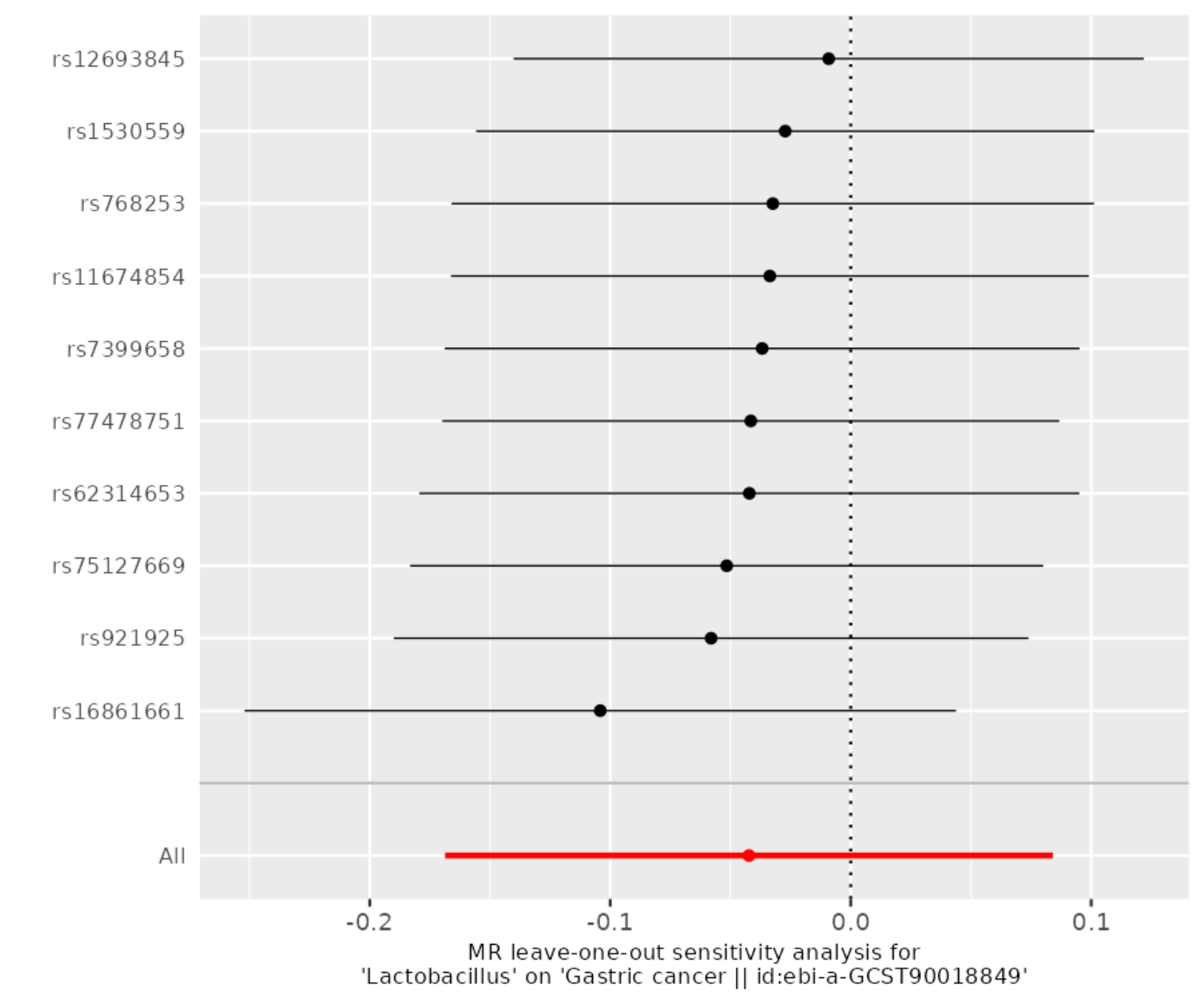
Supplementary Figure S18



Mendelian-randomisation leave-one-out analysis: Veillonella.

results/mr_real/loo_Veillonella.png

Supplementary Figure S19



Mendelian-randomisation leave-one-out analysis: Lactobacillus.

results/mr_real/loo_Lactobacillus.png